

**GSM BSS
GBSS26.1**

Dynamic Channel Conversion Feature Parameter Description

Issue 03
Date 2025-12-20



Copyright © Huawei Technologies Co., Ltd. 2026. All rights reserved.

No part of this document may be reproduced or transmitted in any form or by any means without prior written consent of Huawei Technologies Co., Ltd.

Trademarks and Permissions



HUAWEI and other Huawei trademarks are trademarks of Huawei Technologies Co., Ltd.

All other trademarks and trade names mentioned in this document are the property of their respective holders.

Notice

The purchased products, services and features are stipulated by the contract made between Huawei and the customer. All or part of the products, services and features described in this document may not be within the purchase scope or the usage scope. Unless otherwise specified in the contract, all statements, information, and recommendations in this document are provided "AS IS" without warranties, guarantees or representations of any kind, either express or implied.

The information in this document is subject to change without notice. Every effort has been made in the preparation of this document to ensure accuracy of the contents, but all statements, information, and recommendations in this document do not constitute a warranty of any kind, express or implied.

Huawei Technologies Co., Ltd.

Address: Huawei Industrial Base
Bantian, Longgang
Shenzhen 518129
People's Republic of China

Website: <https://www.huawei.com>

Email: support@huawei.com

Security Declaration

Vulnerability

Huawei's regulations on product vulnerability management are subject to the *Vul. Response Process*. For details about this process, visit the following web page:

<https://www.huawei.com/en/psirt/vul-response-process>

For vulnerability information, enterprise customers can visit the following web page:

<https://securitybulletin.huawei.com/enterprise/en/security-advisory>

Contents

1 About This Document.....**1**

1.1 Scope..... 1

1.2 General Statements..... 1

1.3 Change History..... 2

2 Overview.....**4**

2.1 Introduction..... 4

2.2 NEs Supporting the Feature..... 4

3 Technical Description.....**6**

3.1 SDCCH Dynamic Adjustment..... 6

3.1.1 Overview..... 6

3.1.2 Dynamic SDCCH Conversion Decision..... 7

3.1.3 Dynamic SDCCH Conversion Execution..... 12

3.1.4 SDCCH Reversion Decision..... 13

3.1.5 SDCCH Reversion Procedure..... 14

3.1.6 Impact..... 14

3.2 Dynamic Adjustment Between FR and HR..... 15

3.2.1 Overview..... 15

3.2.2 Dynamic TCH HR/FR Conversion During Channel Allocation..... 15

3.2.2.1 Overview..... 15

3.2.2.2 Rate Adjustment Algorithm..... 15

3.2.2.3 Procedures..... 19

3.2.3 Dynamic TCH HR/FR Conversion During a Call..... 21

3.2.3.1 Overview..... 21

3.2.3.2 Rate Adjustment Algorithm..... 22

3.2.3.3 Procedures..... 25

3.3 Dynamic PDCH Conversion..... 25

3.3.1 Overview..... 25

3.3.2 Dynamic PDCH Conversion Procedure..... 26

3.4 Dynamic Multiple CCCH..... 29

3.4.1 Overview..... 29

3.4.2 Application Scenario..... 31

3.4.3 Adding CCCHs Dynamically..... 32

3.4.4 Deleting CCCHs Dynamically.....	32
4 Related Features.....	33
4.1 GBFD-113001 SDCCH Dynamic Adjustment.....	33
4.2 GBFD-113402 Dynamic Adjustment Between FR and HR.....	34
4.3 GBFD-511503 Dynamic Multiple CCCH.....	34
5 Network Impact.....	36
5.1 Dynamic SDCCH Conversion.....	36
5.1.1 System Capacity.....	36
5.1.2 Network Performance.....	36
5.2 Dynamic TCH HR/FR Conversion.....	36
5.2.1 System Capacity.....	36
5.2.2 Network Performance.....	36
5.3 Dynamic PDCH Conversion.....	37
5.3.1 System Capacity.....	37
5.3.2 Network Performance.....	37
5.4 Dynamic Multiple CCCH.....	37
5.4.1 System Capacity.....	37
5.4.2 Network Performance.....	37
6 Engineering Guidelines.....	44
6.1 When to Use Dynamic Channel Conversion.....	44
6.1.1 Dynamic SDCCH Conversion.....	44
6.1.2 Dynamic TCH HR/FR Conversion.....	44
6.1.3 Dynamic PDCH Conversion.....	45
6.1.4 Dynamic Multiple CCCH.....	45
6.2 Required Information.....	45
6.3 Planning.....	45
6.4 Overall Deployment Procedure.....	46
6.5 Deploying Dynamic SDCCH Conversion.....	47
6.5.1 Deployment Requirements.....	47
6.5.2 Activation.....	47
6.5.3 Activation Observation.....	50
6.5.4 Deactivation.....	50
6.6 Deploying Dynamic TCH HRFR Conversion.....	51
6.6.1 Deployment Requirements.....	51
6.6.2 Dynamic TCH HR/FR Conversion During Channel Allocation.....	51
6.6.2.1 Data Preparation.....	51
6.6.2.2 Activation.....	55
6.6.2.3 Activation Observation.....	59
6.6.2.4 Deactivation.....	60
6.6.3 Dynamic TCH HR/FR Conversion During a Call.....	60
6.6.3.1 Activation.....	60

6.6.3.2 Activation Observation..... 65

6.6.3.3 Deactivation..... 66

6.7 Deploying Dynamic Multiple CCCH..... 67

6.7.1 Deployment Requirements..... 67

6.7.2 Precautions..... 68

6.7.3 Data Preparation..... 68

6.7.4 Activation..... 76

6.7.5 Activation Observation..... 78

6.7.6 Deactivation..... 80

6.8 Performance Optimization..... 80

6.8.1 Dynamic SDCCH Conversion..... 81

6.8.2 Dynamic TCH HR/FR Conversion..... 81

6.8.3 Dynamic PDCH Conversion..... 82

6.8.4 Dynamic Multiple CCCH..... 82

6.9 Troubleshooting..... 83

7 Parameters..... 84

8 Counters..... 86

9 Glossary..... 87

10 Reference Documents..... 88

1 About This Document

1.1 Scope

This document describes Dynamic Channel Conversion, including its technical principles, related features, network impact, and engineering guidelines.

This document covers the following features:

- GBFD-113001 SDCCH Dynamic Adjustment
- GBFD-113402 Dynamic Adjustment Between FR and HR
- GBFD-511503 Dynamic Multiple CCCH

1.2 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements of the operating environment that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

The product supports only the functions mentioned in this document, and the functions can be enabled only in the specified scenarios. If a function not mentioned in this document is enabled or a function is enabled in an unspecified scenario, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

1.3 Change History

This section provides information about the changes in different document versions. There are two types of changes, which are defined as follows:

- Feature change
Changes in features of a specific product version
- Editorial change
Changes in wording or addition of information that was not described in the earlier version

03 (2025-12-20)

This issue includes the following changes.

Change Type	Change Description	Parameter Change
Feature change	None	None
Editorial change	Revised some descriptions in this document, which do not involve changes in meanings and therefore are not detailed.	None

02 (2025-09-26)

This issue includes the following changes.

Change Type	Change Description	Parameter Change
Feature change	None	None
Editorial change	Optimized the description in 3.2.3.2 Rate Adjustment Algorithm .	None

01 (2025-03-28)

GBSS26.1 01 (2025-03-28) introduces the following changes to GBSS24.1 02 (2023-04-23).

Change Type	Change Description	Parameter Change
Feature change	None	None
Editorial change	Optimized the description in 6.5.2 Activation .	None

2 Overview

2.1 Introduction

The Dynamic Channel Conversion feature enables the conversion of a channel from one type to another based on the actual service requirement in the network. Such a conversion is temporary and triggered by a combination of certain network conditions. The converted channel reverts to the original type when the triggering conditions disappear.

Dynamic Channel Conversion includes the following types of conversions:

- Between the traffic channel (TCH) and the standalone dedicated control channel (SDCCH)
- Between the full-rate TCH (TCHF) and the half-rate TCH (TCHH)
- Between the TCH and the packet data channel (PDCH)
- From the dynamic PDCH to the SDCCH
- Dynamic CCCH conversion

Dynamic Channel Conversion facilitates channel configuration for cells and improves the service processing capability of cells.

2.2 NEs Supporting the Feature

Table 2-1 NEs supporting the feature

Feature	BSC6900	BSC6910	GBTS	eGBTS
Dynamic Adjustment Between FR and HR	√	√	√	√
Dynamic Multiple CCCH	√	√	√	√

Feature	BSC6900	BSC6910	GBTS	eGBTS
SDCCH Dynamic Adjustment	√	√	√	√

 NOTE

√ indicates that the NE supports this feature. × indicates that the NE does not support this feature.

3 Technical Description

3.1 SDCCH Dynamic Adjustment

3.1.1 Overview

This section describes the GBFD-113001 SDCCH Dynamic Adjustment feature.

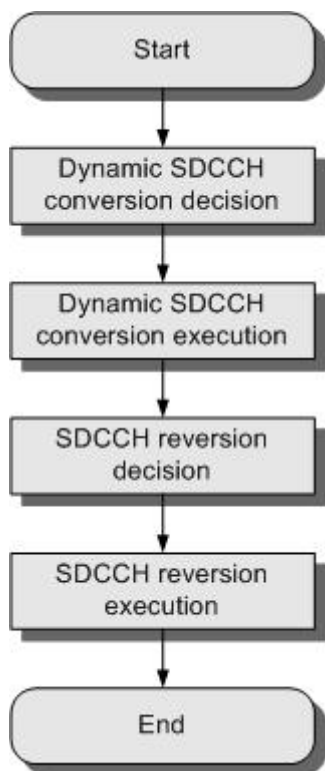
Dynamic SDCCH conversion is triggered when the requested SDCCH is not allocated to the MS. This procedure involves converting a TCHF to the SDCCH, reverting the SDCCH to the TCHF, and converting the dynamic PDCH to SDCCH. (GBFD-113001 SDCCH Dynamic Adjustment)

Dynamic SDCCH conversion helps to decrease the SDCCH congestion rate in the network and reduce the impact of the initial SDCCH configuration on the system performance. Therefore, it reduces the requirement of evaluating the SDCCH demands accurately.

The process of converting the TCHF to the SDCCH and converting the dynamic PDCH to SDCCH consists of two phases: SDCCH conversion decision and SDCCH conversion execution.

An SDCCH can be reverted to a TCHF only. The process of reverting the SDCCH to the TCHF also consists of two phases: SDCCH reversion decision and SDCCH reversion execution. [Figure 3-1](#) shows the complete procedure of dynamic SDCCH conversion.

Figure 3-1 Dynamic SDCCH conversion procedure



The BSC preferentially converts an idle TCHF to an SDCCH. If there is no idle TCHF and the ***PDCH2SDEN (BSC6900, BSC6910)*** parameter is set to **YES(Yes)**, the BSC converts a PDCH to the SDCCH. The PDCH to be converted to the SDCCH must meet the same requirements as the TCHF. This document uses a TCHF as an example for your reference only.

3.1.2 Dynamic SDCCH Conversion Decision

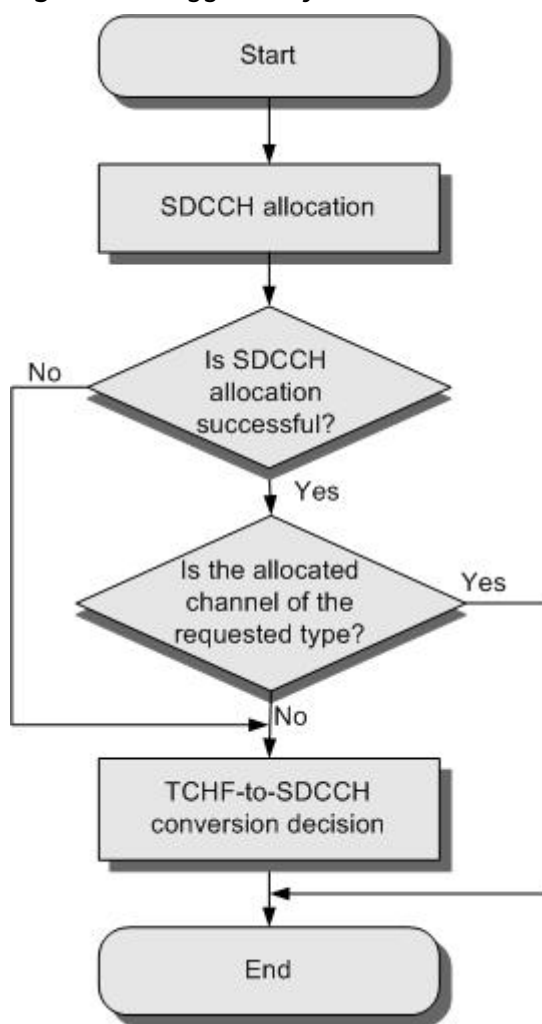
The typical decision-making process for dynamic SDCCH conversion consists of three procedures: trigger of the conversion, decision of the conversion, and TCHF channel selection.

Dynamic SDCCH conversion can be triggered if the number of idle SDCCHs in the cell is smaller than or equal to the value of ***IDLESPTHRES (BSC6900, BSC6910)*** or the SDCCH allocation fails during the channel assignment. If the TCHF is converted to the SDCCH after the SDCCH allocation failure, the converted SDCCH is used for the next SDCCH request.

Trigger of Dynamic SDCCH Conversion

Figure 3-2 shows the trigger process of dynamic SDCCH conversion.

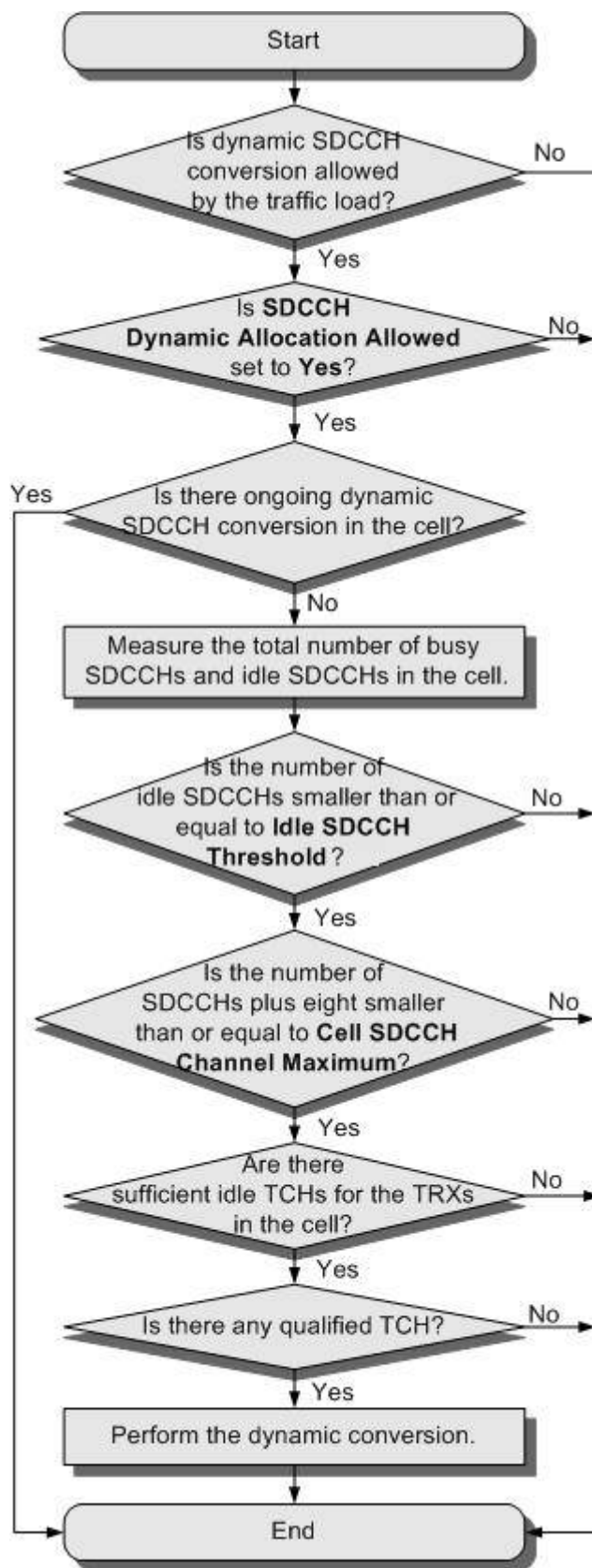
Figure 3-2 Trigger of dynamic SDCCH conversion



Dynamic SDCCH Conversion Decision

Figure 3-3 shows the decision process of dynamic SDCCH conversion.

Figure 3-3 Decision of dynamic SDCCH conversion



After the dynamic SDCCH conversion decision is triggered, the BSC determines whether to allow the conversion based on the following rules.

1. Whether the BSC internal flow control level is 0
If yes, dynamic SDCCH conversion is allowed. Otherwise, it is not allowed.
The BSC internal flow control is performed based on the CPU usage of the current cell. If the CPU usage is less than 65% and no emergency event (for example, massive maintenance operations) occurs, the flow control is not performed, that is, the internal flow control level is 0.
2. Whether **SDDYN (BSC6900, BSC6910)** is set to **Yes**
If yes, dynamic SDCCH conversion is allowed. Otherwise, it is not allowed.
3. Whether there is another ongoing dynamic SDCCH conversion procedure in the cell
If yes, the new conversion procedure is not allowed. Otherwise, it is allowed.

 NOTE

Only one dynamic SDCCH conversion procedure can be performed at a time, in a cell.

4. Whether the number of idle SDCCHs in the cell is smaller than or equal to the value of **IDLESPTHRES (BSC6900, BSC6910)**
If yes, dynamic SDCCH conversion is allowed. Otherwise, it is not allowed.
5. Whether the number of SDCCHs in the cell plus eight is smaller than or equal to the value of **CELLMAXSD (BSC6900, BSC6910)**
If yes, dynamic SDCCH conversion is allowed. Otherwise, it is not allowed.

 NOTE

One TCHF can be converted into eight SDCCHs. Therefore, the total number of SDCCHs after a conversion equals the number of current SDCCHs plus eight.

6. Whether there are sufficient idle TCHs for the TRXs in the cell
If yes, dynamic SDCCH conversion is allowed. Otherwise, it is not allowed.
The decision procedure is as follows:
 - a. The BSC measures the number of idle TCHFs and the number of idle TCHHs in the cell.
 - b. If the **SDDynAdjRsvTchSwitch (BSC6900, BSC6910)** parameter is set to **ON(On)**, the BSC proceeds with [6.5](#).
 - c. The BSC determines whether the number of idle TCHHs plus twice the number of idle TCHFs is greater than eight.
If yes, dynamic SDCCH conversion is allowed. Otherwise, the BSC proceeds with [6.4](#).
 - d. The BSC determines whether the number of idle TCHHs plus twice the number of idle TCHFs is greater than twice the number of TRXs in the cell.
If yes, dynamic SDCCH conversion is allowed. Otherwise, dynamic SDCCH conversion is not allowed.
 - e. The dynamic SDCCH conversion is not allowed if the number of idle TCHHs plus twice the number of idle TCHFs is less than or equal to **SDDynAdjRsvTchNum (BSC6900, BSC6910)**.

7. Whether there is any qualified TCHF for conversion

If yes, dynamic SDCCH conversion is allowed. Otherwise, dynamic SDCCH conversion is not allowed.

TCHF Selection

After the BSC determines that dynamic SDCCH conversion can be performed, it must select a TCHF for conversion.

A qualified TCHF that can be converted to the SDCCH must meet the following requirements:

- The channel must be in the available state.
- If ***DIFFBANDSDCCHDYNADJ (BSC6900, BSC6910)*** is set to **YES(Yes)**, a TCHF carried on a TRX incompatible with the BCCH frequency band can be dynamically converted to an SDCCH. You can set the number of the timeslots on each TRX that can be converted into SDCCHs by setting the ***SDCCHDYNADJTSNUM (BSC6900, BSC6910)*** parameter. If ***DIFFBANDSDCCHDYNADJ (BSC6900, BSC6910)*** is set to **NO(No)**, only a TCHF on the BCCH TRX or a TRX compatible with the BCCH frequency band can be dynamically converted to an SDCCH. The ***MainBcchSdcchNum (BSC6900, BSC6910)*** parameter specifies the number of timeslots corresponding to the TCHFs that can be converted to SDCCHs. The TCHF frequency band and the corresponding BCCH frequency band must be mapped as listed in [Table 3-1](#).
- The channel must be an initial TCHF or a TCHF that is converted from TCHHs.
- The number of existing SDCCHs on the TRX carrying the TCHF must be less or equal than eight x ***SDCCHDYNADJTSNUM (BSC6900, BSC6910)***.

 NOTE

If the channel type of a timeslot is SDCCH/8, the timeslot contains eight SDCCHs.

- The current cell must have more than one TRX.

Table 3-1 Mapping between the BCCH frequency band and the TCHF frequency band for Dynamic SDCCH Conversion

BCCH Frequency Band	TCHF Frequency Band for Dynamic SDCCH Conversion
P-GSM900 M	P-GSM900 M
E-GSM900 M	P-GSM900 M or E-GSM900 M
R-GSM900 M	P-GSM900 M, E-GSM900 M, or R-GSM900 M
DCS1800 M	DCS1800 M
PCS1900 M	PCS1900 M
GSM850 M	GSM850 M
GSM480 M	GSM480 M

BCCH Frequency Band	TCHF Frequency Band for Dynamic SDCCH Conversion
GSM450 M	GSM450 M
TGSM810	TGSM810

The TCHF selection varies with the cell type.

If...	Then...
The cell is a common cell	The BSC selects any of the non-BCCH qualified TCHFs for dynamic SDCCH conversion. NOTE If the serving cell is a single-frequency cell, the BSC selects a BCCH qualified TCHF for dynamic SDCCH conversion.
The cell is a concentric cell	The BSC selects a qualified TCHF in a prioritized order as follows: idle TCHF in the underlaid subcell; busy TCHF in the underlaid subcell; idle TCHF in the overlaid subcell; and busy TCHF in the overlaid subcell.
The cell is a co-BCCH cell	The BSC preferentially selects a qualified idle TCHF in the underlaid subcell. If such an idle TCHF is not found, the BSC selects a qualified busy TCHF. NOTE The TCHFs in the overlaid subcell of a co-BCCH cell cannot be used for dynamic SDCCH conversion.

3.1.3 Dynamic SDCCH Conversion Execution

After selecting a TCHF, the BSC performs the dynamic SDCCH conversion procedure. The dynamic SDCCH conversion procedure varies depending on the status of the selected TCHF.

- If the selected TCHF is idle, the procedure is as follows:
 1. The BSC adjusts the channel attribute from TCHF to SDCCH.
 2. The BSC performs different operations based on whether the channel attribute has been successfully adjusted.
 - If yes, the BSC configures a flag indicating that the cell is under dynamic channel conversion.
 - If no, the BSC terminates the dynamic conversion.

3. The BSC sends the BTS a command instructing the BTS to adjust the channel attribute and measures the counter R3516A: Number of Channel Conversions (TCH-SDCCH).
 4. Upon receiving the command, the BTS starts converting the channels. After the channels are converted, the BTS sends the Status Change Report to the BSC.
 5. Upon receiving the Status Change Report, the BSC sets the status of the converted SDCCH/8 to idle and deletes the flag. The dynamic TCHF-to-SDCCH conversion is complete.
- If the selected TCHF is busy, the procedure is as follows:
 1. To prevent the services carried on the selected TCHF from being interrupted, the BSC initiates a forced intra-cell handover.
 2. The BSC sets the flag indicating that the cell is under dynamic channel conversion and sets the channel status to "TCHF converted to SDCCH, waiting for forced handover response" to prevent the TCHF from being assigned to other services.
 3. The BSC performs different operations based on whether a forced handover response is received.
 - If the BSC receives the forced handover response, it determines whether the forced handover is successful.
 - If the forced handover fails, the TCHF is not released and the BSC deletes the dynamic conversion ongoing flag of the cell. The channel conversion is terminated.
 - If the forced handover is successful, the BSC sets the channel status to "Waiting for channel release". When the TCHF becomes idle, the BSC follows the procedure for converting the idle TCHF to the SDCCH.
 - If the BSC does not receive any forced handover response, the BSC deletes the flag. The channel conversion is terminated.

 NOTE

The TCH far from the PDTCH is preferentially selected for dynamic SDCCH conversion.

3.1.4 SDCCH Reversion Decision

After the TCHF is converted to the SDCCH, the SDCCH is reverted to the TCHF if certain conditions are met.

The BSC determines to revert the SDCCH to the TCHF if the following conditions are met:

- The period in which a TCH or dynamic PDCH is converted to an SDCCH and then reverted to a TCH is longer than or equal to **MINRESTIMETCH** (**BSC6900**, **BSC6910**).
- The number of idle SDCCHs must be greater than the value of **IDLESDTHRES** (**BSC6900**, **BSC6910**) plus 8.
- The selected SDCCH/8 channels are converted from a TCHF.
- One or more of the selected SDCCH/8 channels are idle.
- The channel selection varies with the type of the cell:

- If the cell is not a concentric cell, the SDCCH reversion can be performed when the previous conditions are met.
- If the cell is a concentric cell, the BSC preferentially selects an SDCCH that meets the previous conditions on the TRX in the overlaid subcell. If there is no appropriate SDCCH in the overlaid subcell, the BSC selects an appropriate SDCCH in the underlaid subcell.

If the above conditions are not met, the reversion from the SDCCH to the TCHF is not triggered.

3.1.5 SDCCH Reversion Procedure

If the requirements for SDCCH reversion are met, the BSC performs the SDCCH reversion procedure.

The SDCCH reversion procedure lasts less than three seconds. A busy SDCCH/8 channel can also be reverted to the TCHF. Therefore, based on the status of the selected SDCCH/8 channels, the SDCCH reversion procedure is classified into the following types:

If all the selected SDCCH/8 channels are idle, the SDCCH reversion procedure is as follows:

1. The BSC adjusts the channel attribute from SDCCH to TCHF.
2. The BSC sends a command to the BTS for adjusting the channel attribute and starts the relevant performance measurement.
3. Upon receiving the command, the BTS starts converting the channels. After the channels are converted, the BTS sends the Status Change Report to the BSC.
4. Upon receiving the Status Change Report, the BSC sets the status of the converted TCHF to idle. The SDCCH-to-TCHF reversion is complete.

If one or more of the selected SDCCH/8 channels are busy, the SDCCH reversion procedure is as follows:

1. To prevent call disruption on the selected SDCCH/8 channel, the BSC triggers the timer, waiting for the release of channels.
2. The BSC sets the channel status to "SDCCH/8 converted to TCHF, waiting for idle channel" to prevent these channels from being assigned before reversion.
3. If the sub-channels are still busy when the timer expires, they are released forcibly.
4. When all the selected SDCCH/8 channels are idle, the BSC follows the related procedure for reverting the idle SDCCH.

NOTE

Typically, all the SDCCH/8 channels will be idle before the timer expires, because the SDCCH is occupied for a short time.

3.1.6 Impact

The dynamic SDCCH conversion affects the system performance mainly in the following aspects:

- The bearer capacity of TCHs in a cell is reduced because TCHFs are converted to SDCCHs.
- The intra-cell handovers are increased, if busy TCHFs are converted to SDCCHs.

3.2 Dynamic Adjustment Between FR and HR

3.2.1 Overview

The dynamic TCH HR/FR conversion (GBFD-113402 Dynamic Adjustment Between FR and HR) consists of the following aspects:

- Dynamic TCH HR/FR conversion during channel allocation
During channel allocation, the BSC adjusts the TCHs allocated to MSs in real time based on the cell load and requested TCH type.
- Dynamic TCH HR/FR conversion during a call
During a call, the BSC dynamically adjusts TCHs based on the cell load and receive quality of MSs.

3.2.2 Dynamic TCH HR/FR Conversion During Channel Allocation

3.2.2.1 Overview

This section describes the GBFD-113402 Dynamic Adjustment Between FR and HR feature.

Dynamic TCH HR/FR conversion is performed in two cases: when the traffic load is high in a cell, certain full-rate (FR) TCHs are converted to half-rate (HR) TCHs to allow more MSs to access the network; when the traffic load is low, certain TCHHs are converted to TCHF to provide better speech quality to the MS users. This function is performed by using the TCH HR/FR rate adjustment algorithm.

Dynamic TCH HR/FR conversion is performed in real time during the channel allocation. If a connection requires a TCHH but no idle TCHH is available in the cell, the BSC selects a TCHF to convert it into two TCHHs and then allocates one of them to the connection. If a connection requires a TCHF but no idle TCHF is available, the BSC selects a couple channel to convert it into a TCHF and then allocates the TCHF to the connection. If there is no couple channel available in the cell, the BSC selects a single channel, steers the traffic on the peer busy TCHH to another channel, and then combines two idle single channels into a couple channel. Then, the BSC converts the couple channel to a TCHF.

Set ***TCHAJFLAG (BSC6900, BSC6910)*** to **YES(Yes)** to start the TCH HR/FR conversion during channel allocation.

3.2.2.2 Rate Adjustment Algorithm

The rate adjustment algorithm varies with different rate of requested channels. The required channel rate is defined in the channel assignment request as follows:

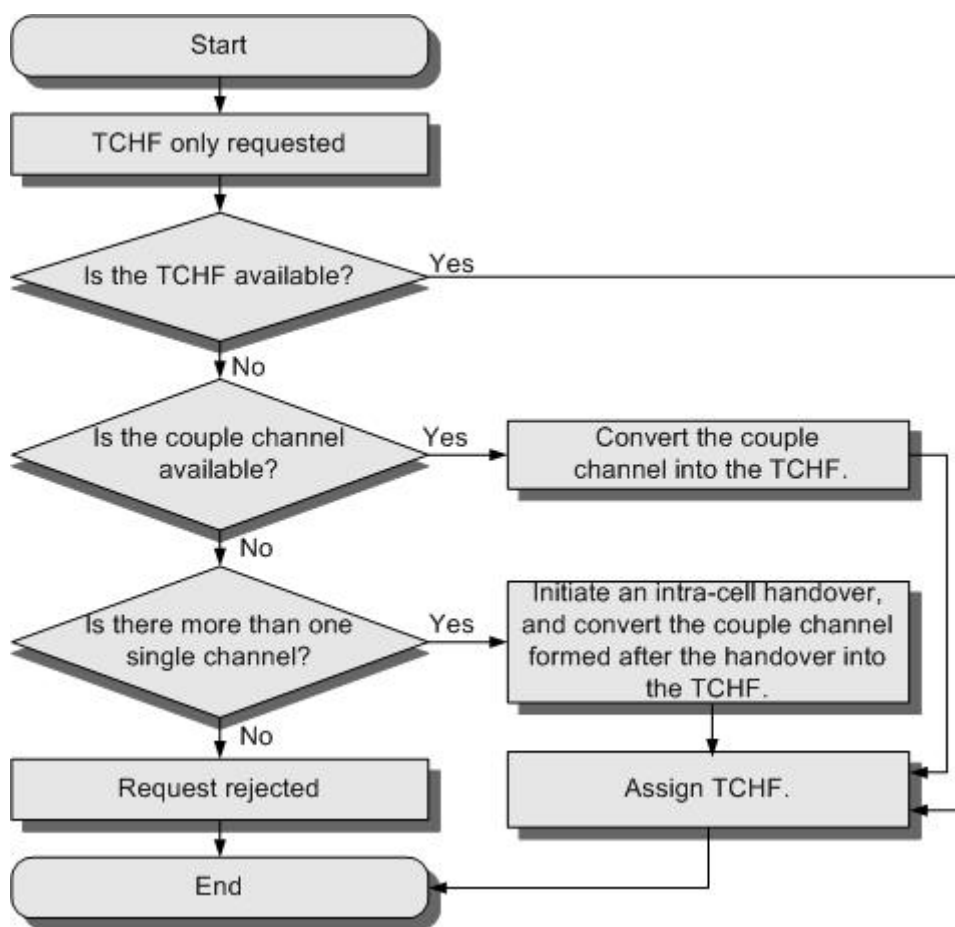
- TCHF only
- TCHH only/TCHH preferred
- TCHF preferred

TCHF only

Figure 3-4 shows the channel assignment procedure when the channel rate type in a channel assignment request is TCHF only. The channel request is accepted in the following situations; otherwise, the channel request is rejected.

- If an idle TCHF is available, the BSC assigns the TCHF directly.
- If no idle TCHF is available but an idle couple channel is available (One timeslot carries two TCHHs. If both TCHHs are idle, they are called couple channels.), convert couple channels to TCHFs and allocate the TCHFs.
- If no idle TCHF or idle couple channel is available but an idle single channel is available (One timeslot carries two TCHHs. If one of them is occupied, the idle TCHH is called a single channel, channel fragment, or fragment), the BSC determines whether there is more than one idle and adjustable TCHH in the cell. If there is more than one idle TCHH in the cell, the BSC initiates the intra-cell handover. Then, the BSC converts the couple channel generated from the intra-cell handover to the TCHF and assigns the TCHF accordingly.

Figure 3-4 Channel rate adjustment procedure for TCHF only



TCHH only/TCHH preferred

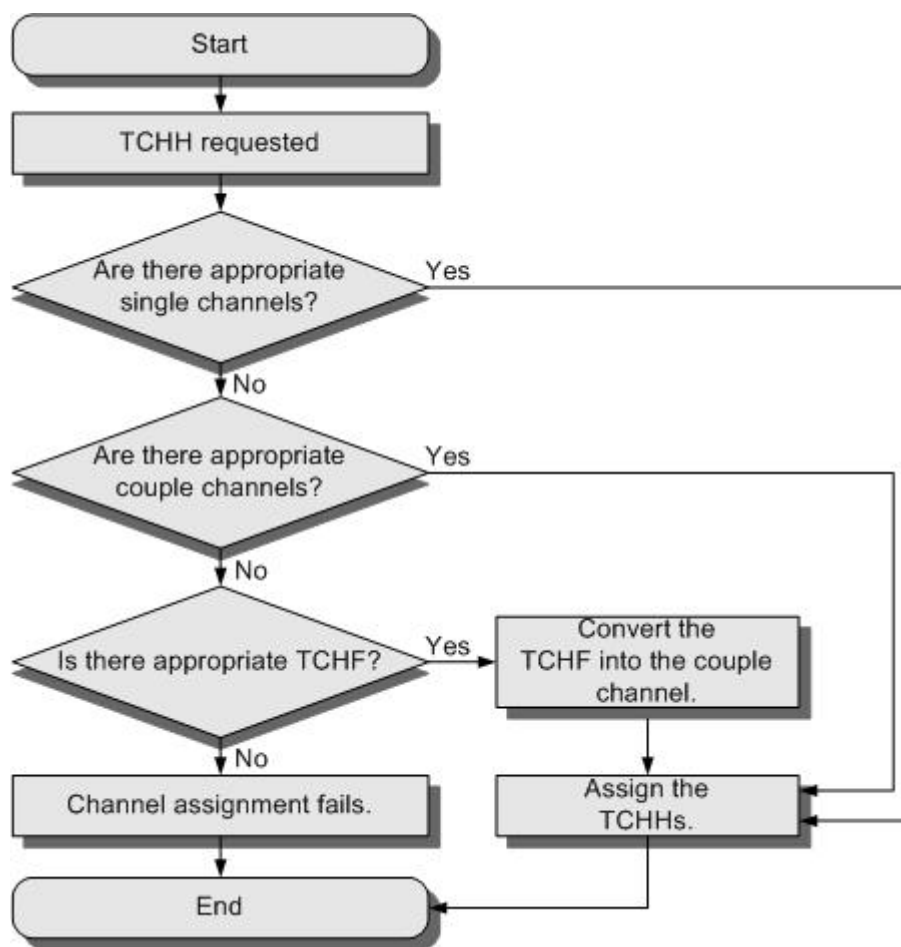
If the channel rate type in a channel assignment request is TCHH only/TCHH preferred, the channel request is accepted in the following situations; otherwise, the channel request is rejected. **Figure 3-5** shows the channel rate adjustment procedure.

- If an idle single channel is available, the BSC assigns the single channel directly.
- If no idle single channel is available but an idle couple channel is available, the BSC assigns the idle TCHH of the couple channel.
- If no idle single channel or couple channel is available but an idle TCHF is available, the BSC converts the TCHF to the couple channel and assigns the idle TCHH of the couple channel.

NOTE

If the BSC assigns single channels preferably, the TCHHs can be utilized optimally with the number of fragments reduced.

Figure 3-5 Channel rate adjustment procedure for TCHH only/TCHH preferred



When the TCHH preferable policy is used and *JudgeRxLevWhenAssignHr* (*BSC6900*, *BSC6910*) is set to **ON(On)**, the BSC determines whether the receive

signal level is lower than a specified threshold. If yes, TCHF are assigned preferentially. If no, TCHH are assigned preferentially. The details are as follows:

For adaptive multirate (AMR) speech services: If the receive signal level is lower than ***MinRxLevWhenAssignHr (BSC6900, BSC6910)***, TCHF are assigned preferentially; otherwise, TCHH are assigned preferentially. For non-AMR speech services:

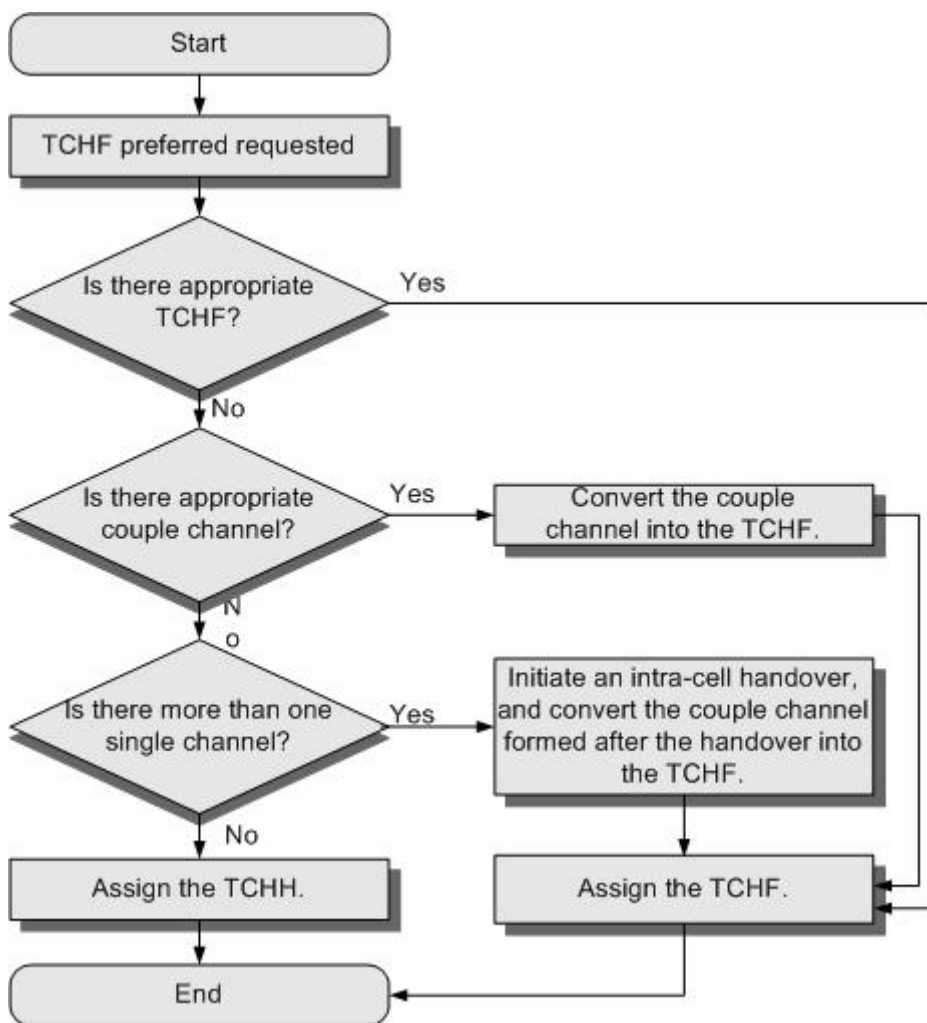
- When ***HOCTRLSWITCH (BSC6900, BSC6910)*** is set to **HOALGORITHM1 (Handover algorithm I)**: If the receive signal level is lower than ***MinRxLevWhenAssignHr (BSC6900, BSC6910)*** plus 5, TCHF are assigned preferentially; otherwise, TCHH are assigned preferentially.
- When ***HOCTRLSWITCH (BSC6900, BSC6910)*** is set to **HOALGORITHM2 (Handover algorithm II)**: If the receive signal level is lower than ***MinRxLevWhenAssignHr (BSC6900, BSC6910)*** + ***(AMRFULLTOHALFHOPATHTHRESH (BSC6900, BSC6910) - NOAMRFULLTOHALFHOPATHTHRESH (BSC6900, BSC6910))***, TCHF are assigned preferentially; otherwise, TCHH are assigned preferentially.

TCHF preferred

If the channel rate type in a channel assignment request is TCHF preferred, the channel request is accepted when the following requirements are met; otherwise, the channel request is rejected. [Figure 3-6](#) shows the channel rate adjustment procedure.

- If an idle TCHF is available, the BSC assigns the TCHF directly.
- If no idle TCHF is available but an idle couple channel is available, the BSC converts the couple channel to the TCHF and then assigns the TCHF.
- If no idle TCHF or couple channel is available but an idle single channel is available, the BSC determines whether there is more than one idle and adjustable TCHH in the cell. If there is more than one idle TCHH in the cell, the BSC initiates the intra-cell handover. Then, the BSC converts the couple channel generated from the intra-cell handover to the TCHF and assigns the TCHF accordingly. Otherwise, the BSC assigns the idle single channel.

Figure 3-6 Channel rate adjustment procedure for TCHF preferred



NOTE

The calls on single channels can be handed over to other channels during the TCHH-to-TCHF conversion only when *ENTCHADJALLOW (BSC6900, BSC6910)* is set to **Yes**.

3.2.2.3 Procedures

This describes the TCHF-to-TCHH conversion and the TCHH-to-TCHF conversion.

TCHF-to-TCHH Conversion

1. If there is no TCHH available for a call that requests a TCHH, the channel request is queued.
 - If the queuing fails, then
 - For the TCHH-only request, a message is returned, indicating that the channel request fails.
 - For the TCHH-preferred request, a less-preferred channel such as the couple channel or TCHF is allocated.
 - If the queuing is successful, then

The BSC converts a TCHF to a couple channel and adjusts the corresponding channel attributes. Then, it issues the BSC timeslot conversion command, and starts the relevant performance measurement.

2. The BSC issues a channel type adjustment command to the BTS. The BSC timeslot adjustment command or the channel type adjustment command will be sent again upon timeout.
3. On receiving the adjustment command, the BTS starts channel type adjustment. Once the channel type adjustment is complete, the BTS sends the Status Change Report to the BSC.
4. On receiving the Status Change Report from the BTS, the BSC sets the couple channel to the idle state. Therefore, the TCHF-to-TCHH conversion is complete. Then, the BSC assigns the converted channel to the call in the queue and starts the measurement of relevant performance statistics.

TCHH-to-TCHF Conversion

1. If a couple channel is available for a call that requests a TCHF, the channel request is queued.
 - If the queuing fails, then
 - For the TCHF-only request, a message is returned, indicating that the channel request fails.
 - For the TCHF-preferred request, a less-preferred channel such as a couple channel or a single channel is allocated.
 - If the queuing is successful, then

The BSC converts the couple timeslots into a TCHF channel. For details about the subsequent procedures, see [TCHF-to-TCHH Conversion](#).
2. If a single channel is available for a call that requests a TCHF, the channel request is queued. The duration for queuing is set to 5.5 seconds, including the time for adjusting the single channels to the couple channel.
 - If the queuing fails, then
 - For the TCHF-only request, a message is returned, indicating that the channel request fails.
 - For the TCHF-preferred request, the single channel is allocated.
 - If the queuing is successful, then

The BSC steers the traffic in the busy peer TCHH channel of the single channel to another channel by initiating the forced intra-cell handover regardless of the settings of handover parameters.

Different operations are performed based on whether a response to the forced handover is received.

 - If the BSC receives the response to the forced handover, it determines whether the forced handover is successful. If the forced handover fails, the BSC allocates the single channel to another call in the queue and does not release the busy peer TCHH channel. If the forced handover is successful, the channel status is set to Waiting for Releasing the Channel; when the channel status is Service Idle, the BSC converts the couple channel into a TCHF.
 - If the BSC does not receive any response to the forced handover, the handover failure procedure is triggered.

3.2.3 Dynamic TCH HR/FR Conversion During a Call

3.2.3.1 Overview

Dynamic TCHH/TCHF conversion during a call enables the BSC to dynamically adjust TCHs based on the cell load or receive quality of MSs during a call.

Handovers between a TCHF and a TCHH are classified into the following types:

- AMR TCHF-to-TCHH handover
- Non-AMR TCHF-to-TCHH handover
These types of handovers enable the BSC to hand over calls from TCHFs to TCHHs when the cell load is high. This increases cell capacity and reduces cell congestion.
- AMR TCHH-to-TCHF handover
- Non-AMR TCHH-to-TCHF handover
These types of handovers enable the BSC to hand over calls from TCHHs to TCHFs when the cell load is low. This improves voice quality.

[Table 3-2](#) lists the algorithm combinations and how to enable them.

Table 3-2 Usage scenarios for handover algorithms

No.	Algorithm Combination	How to Enable Algorithm Combination
1	HUAWEI I Handover and Dynamic HR/FR Adaptation	Set <i>HOCTRLSWITCH (BSC6900, BSC6910)</i> to <i>HOALGORITHM1</i> (Hand over algorithm I), <i>INTRACELLFHHOEN (BSC6900, BSC6910)</i> to <i>HOBASECAPANDQUAL (HOBASECAPANDQUAL)</i> , <i>AMRFULLTOHALFHOALLOW (BSC6900, BSC6910)</i> to YES(Yes), and <i>NOAMRFULLTOHALF-HOALLOW (BSC6900, BSC6910)</i> to YES(Yes).

No.	Algorithm Combination	How to Enable Algorithm Combination
2	HUAWEI II Handover and Dynamic HR/FR Adaptation	Set HOCTRLSWITCH (BSC6900, BSC6910) to HOALGORITHM2 (Hand over algorithm II), AMRFULLTOHALFHOALLOW (BSC6900, BSC6910) to YES(Yes), and NOAMRFULLTOHALFHOALLOW (BSC6900, BSC6910) to YES(Yes).

3.2.3.2 Rate Adjustment Algorithm

Trigger Conditions for an AMR TCHF-TCHH Handover

If **AMRFULLTOHALFHOALLOW (BSC6900, BSC6910)** is set to YES(Yes), the trigger conditions for an AMR TCHF-TCHH handover are as follows:

- **VOICEVER (BSC6900, BSC6910)** used by the call is full-rate speech version 3 or full-rate speech version 5.
- The proportion of MSs occupying TCHHs is less than **ALLOWHALFRATEUSERPERC (BSC6900, BSC6910)**.
- When **F2HHOLOADSTFSWITCH (BSC6900, BSC6910)** is set to ON(On):
 - If the cell load is higher than **AMRFULLTOHALFHOTHRESH (BSC6900, BSC6910)**, the BSC hands over all calls meeting conditions in a cell from TCHFs to TCHHs within the period specified by **FULLTOHALFHODURATION (BSC6900, BSC6910)**.
 - If the cell load is lower than the value of **AMRFULLTOHALFHOTHRESH (BSC6900, BSC6910)** minus **FullToHalfHoLoadStf (BSC6900, BSC6910)**, the BSC stops the TCHF-to-TCHH handover triggered by the cell load.
- When **F2HHOLOADSTFSWITCH (BSC6900, BSC6910)** is set to OFF(Off), if the cell load is higher than **AMRFULLTOHALFHOTHRESH (BSC6900, BSC6910)**, the BSC hands over all calls meeting conditions in a cell from TCHFs to TCHHs within the period specified by **FULLTOHALFHODURATION (BSC6900, BSC6910)**.
- Downlink path loss of the call $\leq$ **AMRFULLTOHALFHOPATHTHRESH (BSC6900, BSC6910)** + Path loss offset
 Path loss offset = (duration for triggering handover from full rate to half rate/ **FULLTOHALFHOPERIOD (BSC6900, BSC6910)** + 1) x **AMRFULLTOHALFHOPATHADJUSTEP (BSC6900, BSC6910)**
- ATCB of the call $\geq$ **AMRFULLTOHALFHOATCBTHRESH (BSC6900, BSC6910)** – ATCB offset

ATCB offset = (duration for triggering handover from full rate to half rate/
FULLTOHALFHOPERIOD (BSC6900, BSC6910) + 1) x
AMRFULLTOHALFHOATCBADJUSTSTEP (BSC6900, BSC6910)

- Uplink signal quality of the call $\leq$ *AMRFULLTOHALFHOQUALTHRESH (BSC6900, BSC6910)*
- Downlink signal quality of the call $\leq$ *AMRFULLTOHALFHOQUALTHRESH (BSC6900, BSC6910)*
- The preceding conditions are met for *FULLTOHALFHOLASTTIME (BSC6900, BSC6910)* within *FULLTOHALFHOSTATTIME (BSC6900, BSC6910)*.

Trigger Conditions for a Non-AMR TCHF-TCHH Handover

If *NOAMRFULLTOHALFHOALLOW (BSC6900, BSC6910)* is set to YES(Yes), the trigger conditions for a non-AMR TCHF-TCHH handover are as follows:

- *VOICEVER (BSC6900, BSC6910)* used by the call is full-rate speech version 1 or full-rate speech version 2.
- The proportion of MSs occupying TCHHs is less than *ALLOWHALFRATEUSERPERC (BSC6900, BSC6910)*.
- When *F2HHOLOADSTFSWITCH (BSC6900, BSC6910)* is set to ON(On):
 - If the cell load is higher than *NOAMRFULLTOHALFTHRESH (BSC6900, BSC6910)*, the BSC hands over all calls meeting conditions in a cell from TCHFs to TCHHs within the period specified by *FULLTOHALFHODURATION (BSC6900, BSC6910)*.
 - If the cell load is lower than the value of *NOAMRFULLTOHALFTHRESH (BSC6900, BSC6910)* minus *FullToHalfHoLoadStf (BSC6900, BSC6910)*, the BSC stops the TCHF-to-TCHH handover triggered by the cell load.
- When *F2HHOLOADSTFSWITCH (BSC6900, BSC6910)* is set to OFF(Off), if the cell load is higher than *NOAMRFULLTOHALFTHRESH (BSC6900, BSC6910)*, the BSC hands over all calls meeting conditions in a cell from TCHFs to TCHHs within the period specified by *FULLTOHALFHODURATION (BSC6900, BSC6910)*.
- Downlink path loss of the call $\leq$ *NOAMRFULLTOHALFHOPATHTHRESH (BSC6900, BSC6910)* + Path loss offset
 Path loss offset = (duration for triggering handover from full rate to half rate/
FULLTOHALFHOPERIOD (BSC6900, BSC6910) + 1) x
NOAMRFULLTOHALFHOPATHADJUSTSTEP (BSC6900, BSC6910)
- ATCB of the call $\geq$ *NOAMRFULLTOHALFHOATCBTHRESH (BSC6900, BSC6910)* – ATCB offset
 ATCB offset = (duration for triggering handover from full rate to half rate/
FULLTOHALFHOPERIOD (BSC6900, BSC6910) + 1) x
NOAMRFULLTOHALFHOATCBADJUSTSTEP (BSC6900, BSC6910)
- Uplink signal quality of the call $\leq$ *NOAMRFULLTOHALFHOQUALTHRESH (BSC6900, BSC6910)*
- Downlink signal quality of the call $\leq$ *NOAMRFULLTOHALFHOQUALTHRESH (BSC6900, BSC6910)*
- The preceding conditions are met for *FULLTOHALFHOLASTTIME (BSC6900, BSC6910)* within *FULLTOHALFHOSTATTIME (BSC6900, BSC6910)*.

Trigger Conditions for an AMR TCHH-TCHF Handover

If **AMRFULLTOHALFHOALLOW** (*BSC6900, BSC6910*) is set to **YES(Yes)** or **AMRHALFTOFULLHOQUALALLOW** (*BSC6900, BSC6910*) is set to **YES(Yes)**, the trigger conditions for an AMR TCHH-TCHF handover are as follows:

- **VOICEVER** (*BSC6900, BSC6910*) used by the call is half-rate speech version 3.
- When **F2HHOLOADSTFSWITCH** (*BSC6900, BSC6910*) is set to **ON(On)**:
 - If the cell load is lower than **AMRHALFTOFULLHOTHRESH** (*BSC6900, BSC6910*), the BSC hands over all calls meeting conditions in a cell from TCHHs to TCHFs within the period specified by **HALFTOFULLHODURATION** (*BSC6900, BSC6910*).
 - If the cell load is higher than the value of **AMRHALFTOFULLHOTHRESH** (*BSC6900, BSC6910*) plus **FullToHalfHoLoadStf** (*BSC6900, BSC6910*), the BSC stops the TCHH-to-TCHF handover triggered by the cell load.
- When **F2HHOLOADSTFSWITCH** (*BSC6900, BSC6910*) is set to **OFF(Off)**, if the cell load is lower than **AMRHALFTOFULLHOTHRESH** (*BSC6900, BSC6910*), the BSC hands over all calls meeting conditions in a cell from TCHHs to TCHFs within the period specified by **HALFTOFULLHODURATION** (*BSC6900, BSC6910*).
- Downlink path loss $\geq$ **AMRHALFTOFULLHOPATHTHRESH** (*BSC6900, BSC6910*)
- ATCB of the call $\leq$ **AMRHALFTOFULLHOATCBTHRESH** (*BSC6900, BSC6910*)
- Uplink signal quality of the call $\geq$ **AMRHALFTOFULLHOQUALTHRESH** (*BSC6900, BSC6910*)
- Downlink signal quality of the call $\geq$ **AMRHALFTOFULLHOQUALTHRESH** (*BSC6900, BSC6910*)
- The AMR TCHH-TCHF handover is triggered if **VOICEVER** (*BSC6900, BSC6910*) is half-rate speech version 3 and any of the preceding conditions is met for **HALFTOFULLHOLASTTIME** (*BSC6900, BSC6910*) within **HALFTOFULLHOSTATTIME** (*BSC6900, BSC6910*).

Trigger Conditions for a Non-AMR TCHH-TCHF Handover

If **NOAMRFULLTOHALFHOALLOW** (*BSC6900, BSC6910*) is set to **YES(Yes)** or **NOAMRHALFTOFULLHOQUALALLOW** (*BSC6900, BSC6910*) is set to **YES(Yes)**, the trigger conditions for a non-AMR TCHH-TCHF handover are as follows:

- **VOICEVER** (*BSC6900, BSC6910*) used by the call is half-rate speech version 1.
- When **F2HHOLOADSTFSWITCH** (*BSC6900, BSC6910*) is set to **ON(On)**:
 - If the cell load is lower than **NOAMRHALFTOFULLTHRESH** (*BSC6900, BSC6910*), the BSC hands over all calls meeting conditions in a cell from TCHHs to TCHFs within the period specified by **HALFTOFULLHODURATION** (*BSC6900, BSC6910*).
 - If the cell load is higher than the value of **NOAMRHALFTOFULLTHRESH** (*BSC6900, BSC6910*) plus **FullToHalfHoLoadStf** (*BSC6900, BSC6910*), the BSC stops the TCHH-to-TCHF handover triggered by the cell load.
- When **F2HHOLOADSTFSWITCH** (*BSC6900, BSC6910*) is set to **OFF(Off)**, if the cell load is lower than **NOAMRHALFTOFULLTHRESH** (*BSC6900, BSC6910*), the BSC hands over all calls meeting conditions in a cell from

TCHHs to TCHF within the period specified by **HALFTOFULLHODURATION** (**BSC6900**, **BSC6910**).

- Downlink path loss of the call $\geq$ **NOAMRHALFTOFULLHOPATHTHRESH** (**BSC6900**, **BSC6910**)
- ATCB of the call $\leq$ **NOAMRHALFTOFULLHOATCBTHRESH** (**BSC6900**, **BSC6910**)
- Uplink signal quality of the call $\geq$ **NOAMRHALFTOFULLHOQUALTHRESH** (**BSC6900**, **BSC6910**)
- Downlink signal quality of the call $\geq$ **NOAMRHALFTOFULLHOQUALTHRESH** (**BSC6900**, **BSC6910**)
- The non-AMR TCHH-TCHF handover is triggered if **VOICEVER** (**BSC6900**, **BSC6910**) is half-rate speech version 1 and any of the preceding conditions is met for **HALFTOFULLHOLASTTIME** (**BSC6900**, **BSC6910**) within **HALFTOFULLHOSTATTIME** (**BSC6900**, **BSC6910**).

TCHH Adjustment

If a large number of MSs occupying TCHHs are available in a cell and TCHHs are not properly distributed, a large number of single channels will exist in the cell. If **HALFRATEREPACKINGSWITCH** (**BSC6900**, **BSC6910**) is set to **YES(Yes)**, the BSC selects an appropriate MS on a single channel (a TCH) and forcibly hands over it to another single channel. This ensures that sufficient couple channels in the cell can be converted to idle TCHFs.

Target Cell Selection

Only the serving cell can be selected as the target cell of the TCHF-TCHH handover.

3.2.3.3 Procedures

The dynamic TCH HR/FR conversion consists of TCHH-to-TCHF conversion and TCHF-to-TCHH conversion. The specific procedure is as follows:

1. The BSC determines whether to send an HR/FR handover request to an MS based on the triggering conditions.
2. The BSC checks the channel resources in the cell:
 - After detecting idle channels in the cell, the BSC sends a handover command to trigger the HR/FR conversion.
 - After detecting no idle channels in the cell and receiving a message, the BSC increments the number of congestions caused by HR/FR handover by one.

3.3 Dynamic PDCH Conversion

3.3.1 Overview

Dynamic PDCH conversion consists of TCH-to-dynamic PDCH conversion, dynamic PDCH-to-TCH conversion, and dynamic PDCH-to-SDCCH conversion, wherein the TCH may be the TCHF or TCHH.

The process of dynamic PDCH conversion consists of two phases: PDCH conversion decision and PDCH conversion execution. See the *GPRS/EGPRS Channel Management Feature Parameter Description* for the detailed information about TCH-to-PDCH conversion. If a TCH request is received when no idle TCH is available and there are dynamic PDCHs converted from the TCH in the cell, the BSC initiates a dynamic PDCH-to-TCH conversion.

The BSC converts a dynamic PDCH to an SDCCH if there is no idle TCHF available. The SDCCH can be reverted to a TCHF only.

This function enables the dynamic balance between the CS load and the PS load, reducing the difficulties in planning the PS capacity of the cell. In addition, it lowers the possibility of PDCH waste and therefore maximizes the utilization of radio channel resources.

3.3.2 Dynamic PDCH Conversion Procedure

Dynamic PDCH conversion includes TCH-to-PDCH conversion, PDCH-to-TCH conversion, and PDCH-to-SDCCH conversion. For details about dynamic PDCH-to-SDCCH conversion, see section [3.1 SDCCH Dynamic Adjustment](#).

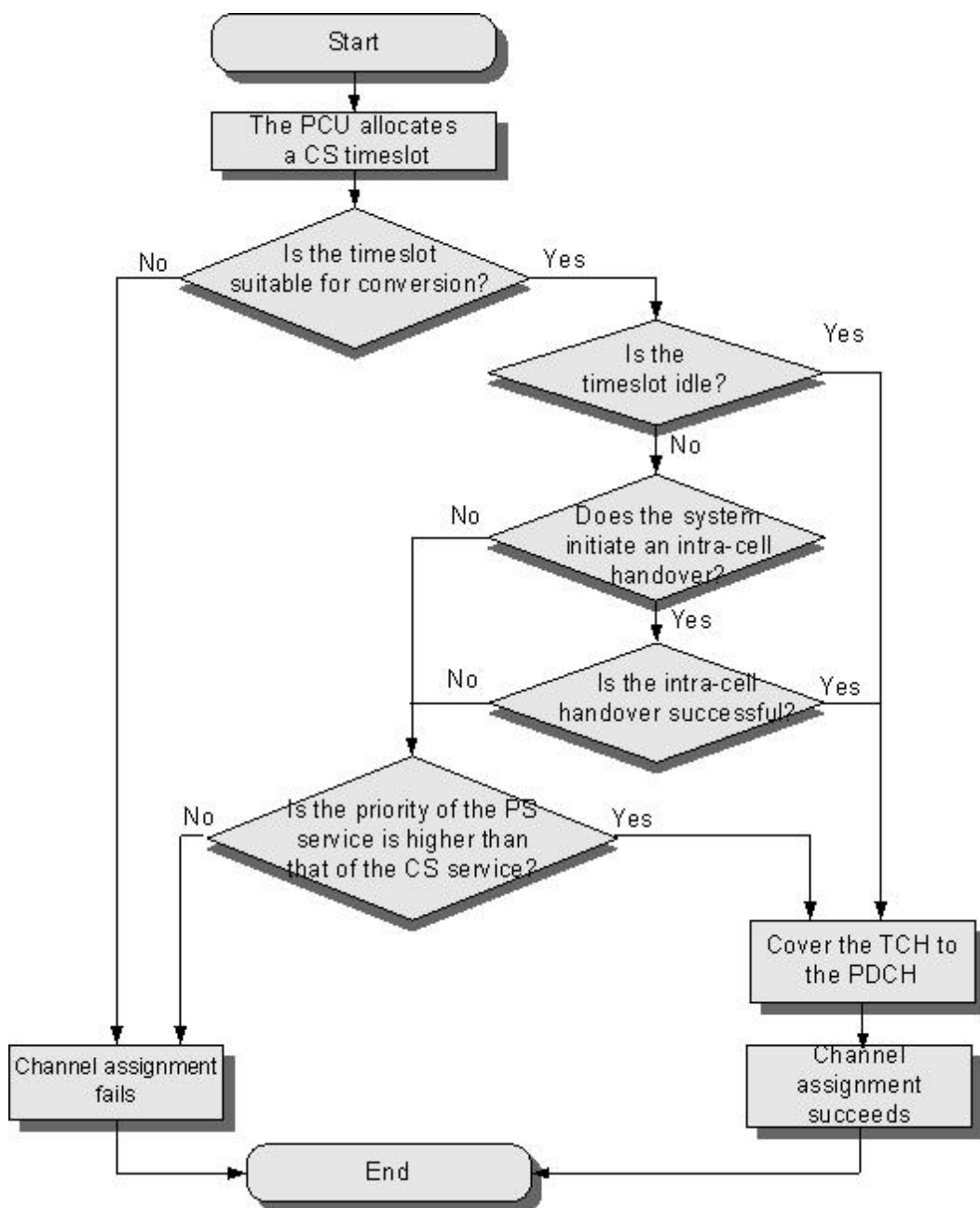
NOTE

The following description is based on the external PCU. If the built-in PCU is configured, then the message interactions between the BSC and the PCU are not needed. The BSC6910 does not support the external PCU.

TCH-to-Dynamic PDCH Conversion

[Figure 3-7](#) shows the TCH-to-dynamic PDCH conversion procedure.

Figure 3-7 TCH-to-dynamic PDCH conversion



1. If there is no PS traffic channel available when an MS initiates the PS services, the PCU allocates a CS timeslot and retrieves the channel type, usage status, and the "whether PDCH assignment request can preempt the channels of CS services" flag of the timeslot.
2. If the **CHTYPE (BSC6900, BSC6910)** of this timeslot is **Dynamic PDCH** and the flag "whether PDCH assignment request can preempt the channels of CS services" is "Yes", the system proceeds with the following procedures:
 - If the timeslot allocated by the PCU is idle, the BSC converts the channel on the timeslot to a PDCH, and then sends a message to the PCU indicating that the channel request is successful.

- If the timeslot (half-rate or full-rate) allocated by the PCU is busy, the BSC determines whether to initiate the intra-cell handover.

The intra-cell handover is not initiated if no idle timeslot is available in the cell or if the channel allocated by the PCU is a TCHH and only one idle timeslot is available in the cell.

If an intra-cell handover cannot be initiated, the BSC determines whether the priority of the PS service is higher than that of the CS service.

- If the priority of the PS service is higher, the BSC releases the channel on the allocated timeslot forcibly and converts the channel to a PDCH.
- If the priority of the CS service is higher, the BSC sends the request failure message to the PCU.

If there are enough idle timeslots in the cell, the BSC initiates an intra-cell handover to steer the traffic on the timeslot to other timeslots.

- If the BSC initiates an intra-cell handover and the handover is successful, the channel on the allocated timeslot is converted to a PDCH.
- If the intra-cell handover fails, the BSC determines whether the priority of the PS service is higher.
- If the priority of the PS service is higher, the BSC releases the channel on the allocated timeslot forcibly and converts the channel to a PDCH.
- If the priority of the CS service is higher, the BSC sends the request failure message to the PCU.

NOTE

To avoid the triggering of a new intra-cell handover, during the channel conversion, the dynamic PDCH is not switched back and the system does not perform half-rate/full-rate adjustment, queuing or preemption.

If the intra-cell handover times out, the system determines whether the priority of the PS service is higher.

- If the priority of the PS service is higher, the BSC releases the channel on the allocated timeslot forcibly and converts the channel to a PDCH.
- If the priority of the CS service is higher, the BSC sends the request failure message to the PCU.

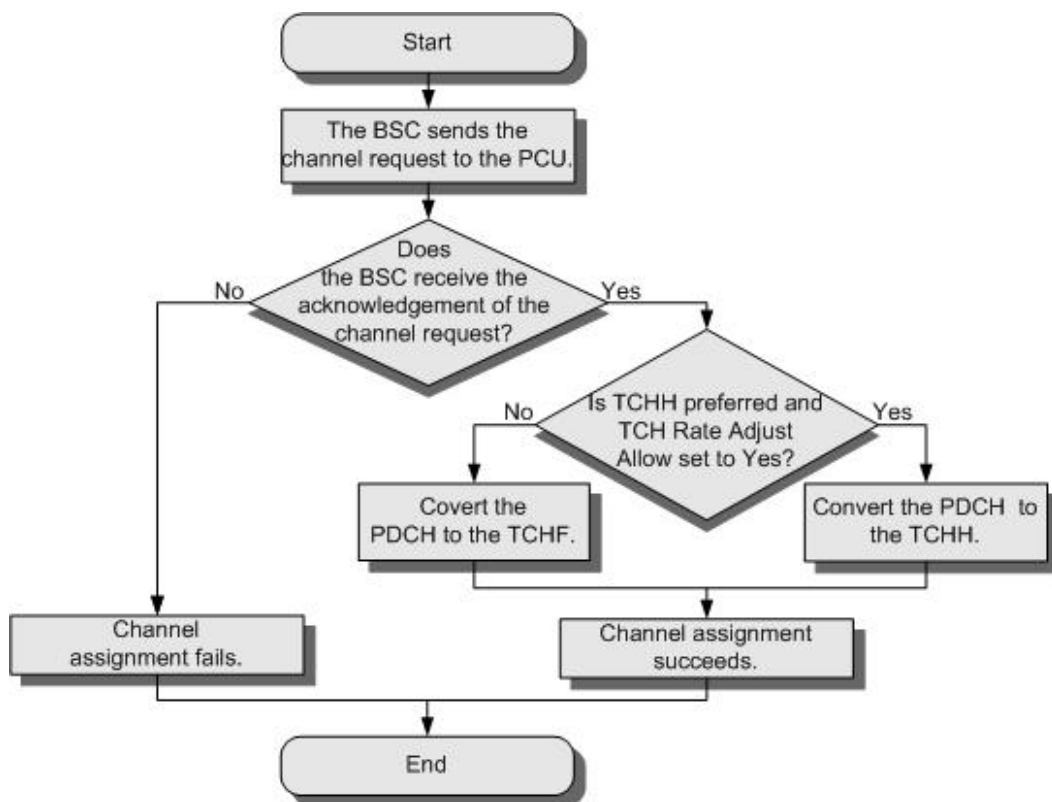
Dynamic PDCH-to-TCH Conversion

NOTICE

- Factors such as the frequency band and concentric cell attribute are considered during dynamic PDCH-to-TCH conversion.
- If the channel requested by the CS call is TCHH only and the TRX carrying the dynamic PDCH to be converted does not support channel rate adjustment, the dynamic PDCH is not converted to the TCH.
- The BSC should not assign the dynamic PDCH for the CS services.

Figure 3-8 shows the dynamic PDCH-to-TCH conversion procedure.

Figure 3-8 Dynamic PDCH-to-TCH conversion



NOTE

- During the dynamic conversion process, the calls are queued. Therefore, if a call is placed in the queue before the BSC sends a channel request message to the PCU, the call will not be released until the queue timer expires.
- If there is no call in the queue when the PCU releases the PDCH, the BSC converts the dynamic PDCH to the TCHF.

When ***TCHBUSYTHRESOPT (BSC6900, BSC6910)*** is set to **OPEN**, the BSC preferentially converts a PDCH to TCHF, and then converts the TCHF to a TCHH. In this case, ***LOADSTATYPE (BSC6900, BSC6910)*** must be set to **Count dynamic PDCH(Occupy)**.

When ***TCHBUSYTHRESOPT (BSC6900, BSC6910)*** is set to **OPEN** and ***LOADSTATYPE (BSC6900, BSC6910)*** is set to **Count dynamic PDCH(Occupy)**, CS services may fail to occupy a dynamic PDCH if ***DYNCHNPREEMPTLEV (BSC6900, BSC6910)*** is set to **LEVEL1(No preempt of CCHs)** or **LEVEL2(No preempt of service TCHF)**. As a result, the KPIs of the CS services will be degraded.

3.4 Dynamic Multiple CCCH

3.4.1 Overview

The paging success rate decreases significantly during holidays or due to emergencies. When the static multiple CCCH function is enabled, it is difficult to

determine the number of static CCCHs to be configured. If the number of configured static CCCHs is small, the paging success rate remains low. If the number of configured static CCCHs is large, some TCHs are wasted.

The GBFD-511503 Dynamic Multiple CCCH feature can be used to resolve this problem. When the CCCH load is high, the TCHs on timeslots 2, 4, and 6 on the BCCH TRX are dynamically converted to CCCHs to extend the Um interface capability. When the CCCH load is low, the CCCHs on timeslots 2, 4, 6 are dynamically converted to TCHs to increase the channel usage.

CCCHs include access grant channels (AGCHs), paging channels (PCHs), and random access channels (RACHs). In the case of traffic burst, the Dynamic Multiple CCCH feature increases the paging success rate, resolves the sharp increase in paging messages, improves the operation and maintenance (O&M) efficiency, and simplifies the multiple CCCH configuration.

After the Dynamic Multiple CCCH feature is enabled, the BTS calculates the CCCH load and determines whether to add or delete CCCHs based on a specified threshold. If yes, the BTS requests the BSC to add or delete CCCHs. The BSC adds or deletes one CCCH per request. When a CCCH is added to a timeslot, the timeslot is chosen by the ascending order of timeslot number. Timeslot 2 is filled before timeslot 4; timeslot 4 is filled before timeslot 6. When a CCCH is deleted from a timeslot, the timeslot is determined by the descending order of the timeslot number. Timeslot 6 is emptied before timeslot 4; timeslot 4 is emptied before timeslot 2.

Channel conversion on timeslots 2, 4, and 6 must be performed consecutively in ascending order. For example, if channel conversion on timeslot 2 fails, channel conversion on timeslots 4 and 6 cannot proceed. After adding or deleting a CCCH, the BSC delivers a system information message to an MS so that the MS listens to the timeslot carrying the allocated CCCH.

To convert a TCH that has calling services, the calling services are handed over to another TCH in the cell. For a common call, if the handover fails, call drops occur. For a VIP call or an emergency call that must be guaranteed, if the handover fails, the call is reassigned to the original channel and therefore call drops do not occur. In this case, a TCH cannot be converted to a CCCH. The VIP calls can be configured by running the **ADD VIPUSR** or **RMV VIPUSR** command. The emergency calls that must be guaranteed can be configured using the **EMCPRILV (BSC6900, BSC6910)** parameter.

To convert a dynamic PDCH, the BSC sends the PCU a request to release this channel. If this channel is successfully released, it is successfully converted to the CCCH. If this channel fails to be released, the channel conversion fails.

The channels to be converted by the Dynamic Multiple CCCH feature cannot be SDCCHs, static PDCHs, channels involved in baseband FH, and channels reserved for VIP users. For details, see [6.3 Planning](#).

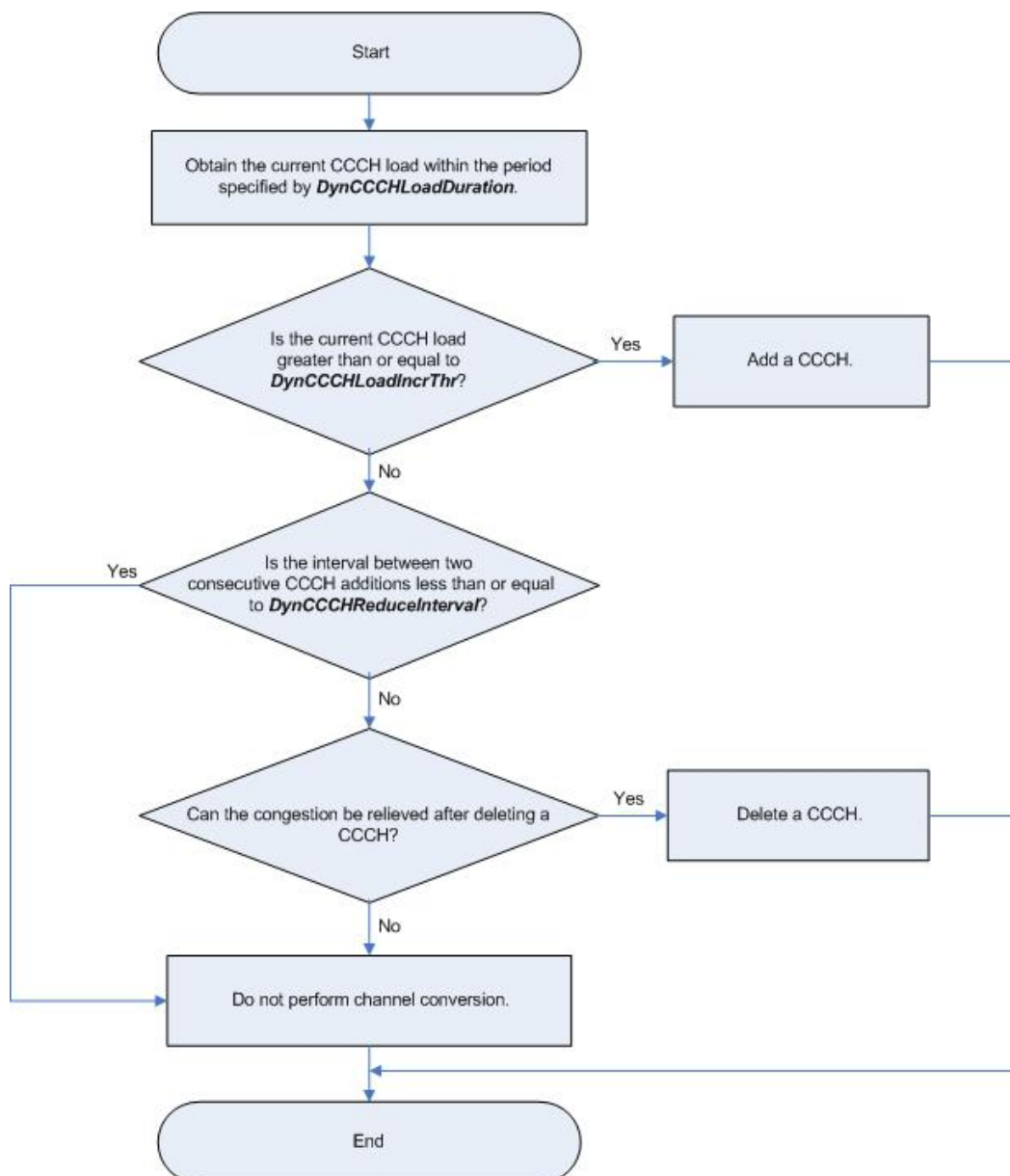
After receiving a request from the BTS to delete a CCCH, the BSC converts the CCCH into a TCH. In addition, the BSC reserves the received paging messages within a specified period, preventing paging message loss caused by channel conversion.

After adding or deleting a CCCH, the BSC sends a System Information 3 (SI3) to notify the MS to listen to the timeslot carrying the CCCH.

Dynamic CCCH conversion is determined by *DynCCCHSwitch* (*BSC6900*, *BSC6910*).

Dynamic CCCH conversion includes adding and deleting CCCHs dynamically. [Figure 3-9](#) shows the dynamic CCCH conversion procedure.

Figure 3-9 Dynamic CCCH conversion procedure



3.4.2 Application Scenario

Huawei BSS device paging methods are classified into local area code (LAC) paging and BSC paging.

Before enabling the Dynamic Multiple CCCH feature, you are advised to disable the configured static broadcast channel (BCH) in a cell.

 NOTE

Multiple CCCHs refer to logical CCCHs on multiple timeslots. Extended CCCHs refer to the CCCHs except the CCCHs on timeslot 0 on the BCCH TRX. For Huawei BSS devices, the channel type for extended CCCHs is BCH, which is the same as the channel type of static CCCHs.

On the live network, if 70% of cells under an LAC or a BSC experience paging expiration during peak hours, you are advised to enable this feature for all the cells under the LAC or BSC.

If the number of delivered paging messages in a cell exceeds 160,000 per hour during peak hours, you are advised to enable this feature for all the cells under the LAC or BSC that serves this cell.

3.4.3 Adding CCCHs Dynamically

Within the period specified by *DynCCCHLoadDuration (BSC6900, BSC6910)*, the BTS measures the CCCH load once every second. If the CCCH load calculated by the BTS is greater than *DynCCCHLoadIncrThr (BSC6900, BSC6910)*, CCCH congestion occurs. Channels on timeslots 2, 4, and 6 of the BCCH TRX are dynamically converted to CCCHs in sequence. If the number of CCCHs plus one is greater than *MaxDynCCCHNum (BSC6900, BSC6910)*, channel conversion is not performed. Otherwise, the BTS sends the BSC a request for applying for a CCCH.

3.4.4 Deleting CCCHs Dynamically

If the interval between two consecutive CCCH additions is less than or equal to *DynCCCHReduceInterval (BSC6900, BSC6910)*, the decision on whether deleting CCCHs is not made. If the interval between two consecutive CCCH additions is greater than *DynCCCHReduceInterval (BSC6900, BSC6910)*, the decision on whether deleting CCCHs is made.

Within the period specified by *DynCCCHLoadDuration (BSC6900, BSC6910)*, the BTS decides once every second whether the following condition is met: Buffer queue length of N CCCHs < Maximum buffer queue length of (N – 1) CCCHs x *DynCCCHLoadRedcThr (BSC6900, BSC6910)*

where N represents the number of CCCH timeslots. N is greater than 1. Maximum buffer queue length of one CCCH = Number of paging groups x Number of buffered CCCHs in each paging group + Allowed length of the AGCH transmission queue

If the preceding condition is met each time the BTS decides during the specified period, the BTS determines whether to delete a CCCH based on the number of downlink messages (including paging messages and immediate assignment messages) over the Abis interface and the capability of transmitting messages (including paging messages and immediate assignment messages) over the Um interface. If the BTS determines to delete a CCCH, it sends the BSC a request for deleting a CCCH dynamically.

4 Related Features

4.1 GBFD-113001 SDCCH Dynamic Adjustment

Prerequisite Features

None.

Mutually Exclusive Features

None.

Impacted Features

Feature ID	Feature Name	Description
GBFD-221201	Dynamic BCCH TRX Timeslot Shutdown	When both Dynamic BCCH TRX Timeslot Shutdown and SDCCH Dynamic Adjustment are enabled, if the channel in a timeslot that is allowed to be shut down is converted to SDCCH, this timeslot cannot be shut down.

4.2 GBFD-113402 Dynamic Adjustment Between FR and HR

Prerequisite Features

Feature ID	Feature Name	Description
GBFD-113401	Half Rate Speech	The dynamic TCH HR/FR conversion during channel allocation depends on this feature.
GBFD-115504	AMR FR/HR Dynamic Adjustment	The dynamic TCH HR/FR conversion during a call depends on either of the features.
GBFD-510501	HUAWEI II Handover	

Mutually Exclusive Features

None.

Impacted Features

Feature ID	Feature Name	Description
GBFD-200201	1.2 MHz Networking for BCCH TRXs	The AMR FR/HR Dynamic Adjustment feature (controlled by the <i>HALFRATEREPACKINGS-WITCH (BSC6900, BSC6910)</i> parameter) and the channel group handover function in the 1.2 MHz Networking for BCCH TRXs feature are mutually impacted. Ping-pong handovers will occur when they are both enabled. You are not advised to enable AMR FR/HR Dynamic Adjustment when 1.2 MHz Networking for BCCH TRXs is enabled.

4.3 GBFD-511503 Dynamic Multiple CCCH

Prerequisite Features

None.

Mutually Exclusive Features

None.

Impacted Features

Feature ID	Feature Name	Description
GBFD-200201	1.2 MHz Networking for BCCH TRXs	When 1.2 MHz Networking for BCCH TRXs is enabled, the transmit power of TS4 on the BCCH TRX will be decreased. You are not advised to configure the BCCH on TS4.
GBFD-221201	Dynamic BCCH TRX Timeslot Shutdown	When both Dynamic BCCH TRX Timeslot Shutdown and Dynamic Multiple CCCH are enabled, if the channel in a timeslot that is allowed to be shut down is converted to CCCH, this timeslot cannot be shut down.

5 Network Impact

5.1 Dynamic SDCCH Conversion

5.1.1 System Capacity

The Dynamic SDCCH Conversion feature increases network signaling traffic and prevents SDCCH congestion due to signaling burst. Examples of SDCCH congestion due to signaling burst are abrupt cell location updates at the border of location areas and group sending of short messages.

After a TCH is successfully converted to SDCCH, the received signal level (RSL) load increases on this channel.

5.1.2 Network Performance

This feature slightly increases the number of handovers.

5.2 Dynamic TCH HR/FR Conversion

5.2.1 System Capacity

When the traffic in a cell is heavy, the TCH HR/FR conversion enables the BSC to allocate TCHHs to MSs or convert the TCHFs to TCHHs. This increases the network capacity. When the traffic in a cell is light, the TCH HR/FR conversion enables the BSC to allocate TCHFs to MSs or convert TCHHs to TCHFs. This improves the voice quality.

5.2.2 Network Performance

The TCH HR/FR conversion prolongs the assignment duration.

In addition, the TCH HR/FR conversion slightly increases the number of handovers and thereby affects the voice quality.

5.3 Dynamic PDCH Conversion

5.3.1 System Capacity

When a cell has heavy CS services, they can occupy PDCHs, improving the CS service capacity. When CS services are light in a cell, PS services can occupy TCHs, improving the PS service throughput.

5.3.2 Network Performance

The PS service rate may decrease or call drops may occur because some PDCHs are occupied by CS services.

5.4 Dynamic Multiple CCCH

5.4.1 System Capacity

The Dynamic Multiple CCCH feature eliminates the avalanche effect caused by paging burst. Compared with the static multiple CCCHs, this feature reduces the operation and maintenance cost and improves the channel usage.

In addition, this feature decreases the number of paging timeouts. When receiving a large number of paging messages, the BSC automatically distributes these messages based on the CCCH load to balance load on CCCHs. This decreases the number of paging timeouts and minimizes the effect caused by paging or PS traffic burst on the paging efficiency.

5.4.2 Network Performance

If the channel to be converted is carrying a PS call, a direct channel preemption is initiated. This may cause the PS call to drop.

Static PDCHs may not be configured in timeslot 6 and PDCH continuity is affected.

If the channel to be converted is carrying a CS call, an intra-cell handover is initiated. This affects the mean opinion score (MOS) of this CS call.

[Table 5-1](#), [Table 5-2](#), and [Table 5-3](#) show the impacts caused by enabling the Dynamic Multiple CCCH feature.

Table 5-1 Counters on the core network side

Counter	Impact
Paging success rate	Slightly increases or decreases. If static BCHs have been configured in a cell and the configuration of the static BCHs is cancelled when the Dynamic Multiple CCCH feature is enabled, some paging messages may be discarded during the triggering of dynamic channel conversion when <i>DynCCCHLoadIncrThr (BSC6900, BSC6910)</i> or <i>DynCCCHLoadDuration (BSC6900, BSC6910)</i> is set to a large value after this feature is enabled. As a result, the paging success rate slightly decreases as compared with the scenario where this feature is not enabled and static BCHs are configured. If no static BCH is configured before the Dynamic Multiple CCCH feature is enabled, this feature increases the paging capability over the air interface. Therefore, the paging success rate increases.

 NOTE

The paging success rate is not a standard counter name on the CN side. It is named based on its meaning. For the specific counter name, see the counter documentation provided by the device provider.

Table 5-2 BSS counters

Type	Counter Description	Counter Name	Impact
Number of paging timeouts	Number of Expiries of the Paging for CS Services on the PCH	CELL.CIRCUIT.PCH.EXP	Slightly increases or decreases. The cause is the same as that described in Table 5-1.
	Number of Expiries of the Paging for PS Services on the PCH	CELL.PACUIT.PCH.EXP	

Type	Counter Description	Counter Name	Impact
CS congestion rate	Congestion Rate on TCH (All Channels Busy)	CELL.KPI.TCH.CONGESTION.RATE	Slightly increases or decreases. If static BCHs have been configured in a cell and the configuration of the static BCHs is cancelled when the Dynamic Multiple CCCH feature is enabled, enabling this feature increases the resource utilization and decreases the congestion rate. If no static BCH is configured when the Dynamic Multiple CCCH feature is enabled, triggering CCCH channel adjustment during peak hours occupies one or more traffic channels after this feature is enabled, leading to a slight increase in the CS congestion rate.

Type	Counter Description	Counter Name	Impact
CS call drop rate	Call Drop Rate on TCH per cell (including Handover)	CELL.RATE.TCH.C ALL.DROP.INCLUD E.HO	Remains unchanged or slightly increases. During a TCH-to-CCCH conversion, if the TCH has been occupied by a CS call, the BSC attempts to hand over the CS call to another channel first. If the handover fails, the BSC directly preempts the TCH. This slightly increases the call drop rate.

Type	Counter Description	Counter Name	Impact
PS service throughput	Average Throughput of Uplink GPRS RLC	UP.GPRS.RLC.AVG.THRPUT.RATE	Increases or slightly decreases. If static BCHs have been configured in a cell and the configuration of the static BCHs is cancelled when the Dynamic Multiple CCCH feature is enabled, enabling this feature increases the resource utilization and the values of these counters. If no static BCH is configured before the Dynamic Multiple CCCH feature is enabled, triggering CCCH channel adjustment during peak hours occupies one or more traffic channels after this feature is enabled, leading to a slight decrease in the PS throughput.
	Average Throughput of Downlink GPRS RLC	DOWN.RLC.AVG.THRPUT.RATE	
	Average Throughput of Uplink EGPRS RLC	UP.EGPRS.RLC.AVG.THRPUT.RATE	
	Average Throughput of Downlink EGPRS RLC	DOWN.EGPRS.RLC.AVG.THRPUT.RATE	

Table 5-3 BSS KPIs

Type	KPI	Impact
Uplink/Downlink GPRS/ EGPRS TBF congestion rate	Uplink TBF Congestion Rate	Decreases or slightly increases.
	Downlink TBF Congestion Rate	If static BCHs have been configured in a cell and the configuration of the static BCHs is cancelled when the Dynamic Multiple CCCH feature is enabled, enabling this feature increases the resource utilization and decreases the values of these counters. If no static BCH is configured before the Dynamic Multiple CCCH feature is enabled, triggering CCCH channel adjustment during peak hours occupies one or more traffic channels after this feature is enabled. As a result, the PS congestion rate slightly increases or keeps unchanged.
	Uplink EGPRS TBF congestion rate	
	Downlink EGPRS TBF congestion rate	
Uplink/Downlink GPRS/ EGPRS TBF call drop rate	Uplink GPRS TBF Call Drop Rate	Remains unchanged or slightly increases.
	Downlink GPRS TBF Call Drop Rate	During the PDCH-to- CCCH conversion, the PCU releases the PDCH first, and then converts the PDCH to a CCCH. If the PDCH is used by a TBF, the TBF release procedure is triggered. This slightly increases the TBF call drop rate.
	Uplink EGPRS TBF Call Drop Rate	
	Downlink EGPRS TBF Call Drop Rate	

Type	KPI	Impact
TBF establishment success rate	Downlink TBF Establishment Success Rate	Remains unchanged or slightly decreases. During the PDCH-to-CCCH conversion, the PCU releases the PDCH first, and then converts the PDCH to a CCCH. This decreases the TBF establishment success rate.
	Uplink TBF Establishment Success Rate	

6 Engineering Guidelines

6.1 When to Use Dynamic Channel Conversion

6.1.1 Dynamic SDCCH Conversion

You are advised to enable the Dynamic SDCCH Conversion feature. To relieve SDCCH congestion, configure certain static SDCCHs in the cells enabled with this feature

You are advised to set ***DIFFBANDSDCCHDYNADJ (BSC6900, BSC6910)*** to **YES(Yes)** if:

- The cell is a co-BCCH cell.
- Some TRXs for a cell whose BCCH TRX is configured on the P-GSM frequency band work on the E-GSM or R-GSM frequency band.
- Only a few TRXs work on the frequency bands compatible with the BCCH frequency band.

NOTICE

During immediate assignment, the BSC cannot determine whether an MS supports a specific frequency band. If an MS does not support a frequency band that is incompatible with the BCCH frequency band but the BSC allocates an SDCCH on such a frequency band to the MS, the MS fails to access the network.

6.1.2 Dynamic TCH HR/FR Conversion

For non-AMR calls, if the cell is a common cell, set ***TCHBUSYTHRES (BSC6900, BSC6910)*** based on your individual needs; If the cell is a concentric cell and ***HRIUOLDRATESELALLOW (BSC6900, BSC6910)*** is set to **YES(Yes)**, set ***TCHTRICBUSYOVERLAYTHR (BSC6900, BSC6910)*** and ***TCHTRIBUSYUNDERLAYTHR (BSC6900, BSC6910)*** based on your individual needs. For AMR calls, in common and concentric cells, set ***AMRTCHHPRIORLOAD (BSC6900, BSC6910)*** based on your individual needs.

If TCHs are assigned preferentially, set ***JudgeRxLevWhenAssignHr (BSC6900, BSC6910)*** and ***MinRxLevWhenAssignHr (BSC6900, BSC6910)*** appropriately to avoid too low signal level.

In channel assignment algorithms, the BSC determines whether to assign TCHs or TCHFs based on the network traffic volume. If the channel usage exceeds ***TCHBUSYTHRES (BSC6900, BSC6910)***, the traffic volume of the current network is high. If the traffic volume of the current network is high, TCHs are assigned preferentially. If the traffic volume of the current network is low, TCHFs are assigned preferentially. If ***JudgeRxLevWhenAssignHr (BSC6900, BSC6910)*** is set to **ON(On)**, TCHs are assigned preferentially when the channel usage exceeds ***TCHBUSYTHRES (BSC6900, BSC6910)*** and the receive signal level is greater than or equal to ***MinRxLevWhenAssignHr (BSC6900, BSC6910)***. Otherwise, TCHFs are assigned preferentially.

It is recommended that the TCH HR/FR conversion be used in cells with heavy traffic and a high proportion of half-rate channels.

6.1.3 Dynamic PDCH Conversion

Before enabling the Dynamic PDCH Conversion feature in a cell, you are advised to configure certain static PDCHs in the cell.

6.1.4 Dynamic Multiple CCCH

None

6.2 Required Information

Before enabling the Dynamic Multiple CCCH feature, check whether all SDCCHs in the cell are configured in timeslots 2, 4, and 6 of the BCCH TRX. If yes, do not enable this feature. If no, obtain the paging success rate in the cell. If the paging success rate is low, enable this feature.

6.3 Planning

Networking planning for the Dynamic Multiple CCCH feature is as follows:

If combined BCCH or BCCH + CBCH is configured in timeslot 0 of the BCCH TRX in a cell, the Dynamic Multiple CCCH feature cannot be enabled.

The following rules are applicable to all the channels involved in baseband FH, the channels configured as SDCCHs or static PDCHs, and the channels reserved for VIP users:

- If ***MaxDynCCCHNum (BSC6900, BSC6910)*** is set to **1**, these channels cannot be configured in timeslot 2. Otherwise, the Dynamic Multiple CCCH feature cannot be enabled.
- If ***MaxDynCCCHNum (BSC6900, BSC6910)*** is set to **2**, these channels cannot be configured in timeslot 2 or timeslot 4. Otherwise, the Dynamic Multiple CCCH feature cannot be enabled.

- If **MaxDynCCCHNum (BSC6900, BSC6910)** is set to **3**, these channels cannot be configured in timeslot 2, timeslot 4, or timeslot 6. Otherwise, the Dynamic Multiple CCCH feature cannot be enabled.

You are advised to set **MaxDynCCCHNum (BSC6900, BSC6910)** to the same value as the number of static CCCHs configured in the cell before this feature is enabled.

If there is no static CCCH in the cell and paging expiration or deletion occurs in 70% of cells under the LAC where this cell is located, you are advised to set **MaxDynCCCHNum (BSC6900, BSC6910)** to **1** or a larger value for all cells under the LAC.

In typical scenarios, the theoretical capability of sending messages over the Um interface per BCCH is calculated using the following formula:

$$(9 - \text{BSAGBLKSRES (BSC6900, BSC6910)}) / 0.235 \times 2.74 \text{ (Unit: integer/s)}$$

NOTE

The typical scenarios must meet the following requirements:

- The first paging is based on the TMSI, while the second paging is based on the IMSI.
- The paging service distribution model is normal. For example, the one-hour model meets the position distribution requirement, but not the M2M. A large number of paging messages are sent within 10s during one hour.

Assume that **BSAGBLKSRES (BSC6900, BSC6910)** is set to **1**, the theoretical capability of sending messages over the Um interface per BCCH is 334800 per hour, which is calculated using the following formula:

$$(9 - 1) / 0.235 \times 2.74 = 93 \text{ (integer/s)}$$

The actual capability, usually, is about 50% to 60% of the theoretical capability. If the number of paging messages sent during peak hours in a cell reaches 160,000 per hour (334800 X 50%), set **MaxDynCCCHNum (BSC6900, BSC6910)** to **1**. If the number of paging messages sent during peak hours in a cell reaches 160,000 x N per hour, set **MaxDynCCCHNum (BSC6900, BSC6910)** to N.

Setting **MaxDynCCCHNum (BSC6900, BSC6910)** to a larger value improves the message transmission capability over the Um interface. To prevent message congestion over the Abis interface, increase the RSL bandwidth for the BCCH TRX and set **MainBMPMode (BSC6900, BSC6910)** as follows:

Temporary mobile subscriber identity (TMSI) paging: When **MaxDynCCCHNum (BSC6900, BSC6910)** plus the number of static BCHs is equal to 1, 2, or 3, you are advised to set **MainBMPMode (BSC6900, BSC6910)** to **MODE2_1**, **MODE1_1**, or **MODE1_1**, respectively.

International mobile subscriber identity (IMSI) paging: When **MaxDynCCCHNum (BSC6900, BSC6910)** plus the number of static BCHs is equal to 1, 2, or 3, you are advised to set **MainBMPMode (BSC6900, BSC6910)** to **MODE4_1**, **MODE2_1**, or **MODE1_1**, respectively.

6.4 Overall Deployment Procedure

None

6.5 Deploying Dynamic SDCCH Conversion

6.5.1 Deployment Requirements

Table 6-1 Deployment requirements

Aspect	Requirement
Related features	See 4 Related Features .
BSC	None
BTS	None
GSM networking	None
MS	None
MSC	None
License	None
Others	None

6.5.2 Activation

Using MML Commands

On the BSC LMT, perform the following steps:

- Step 1** Run the **SET GCELLBASICPARA** command with **SDCCH Dynamic Allocation Allowed** and **Dynamic PDCH Conversion to SDCCH Allowed** set to **YES(Yes)**.
- Step 2** Run the **SET GCELLCHMGBASIC** command with **Idle SDCCH Threshold**, **Cell SDCCH Channel Maximum**, **TCH Minimum Recovery Time**, **Num of TSs Convertible to SDCCHs per TRX**, and **Number of TSs Convertible to SDCCHs on BCCH TRX** set to appropriate values.
-  **NOTE**
- When TCHs on the BCCH frequency band are insufficient, set **Dynamically Adjust Inter-Freq Band SDCCH** to **YES(Yes)**.
- Step 3** Run the **SET GCELLPSCHM** command with **Maximum Rate Threshold of PDCHs in a Cell** and **Reservation Threshold of Dynamic Channel Conversion** set to appropriate values based on the network planning.

----End

MML Command Examples

SET GCELLBASICPARA: IDTYPE=BYID, CELLID=1, SDDYN=YES, PDCH2SDEN=YES;

```
SET GCELLCHMGBASIC: IDTYPE=BYID, CELLID=1, IDLESPTHRES=17,  
CELLMAXSD=100, MINRESTIMETCH=60, SDCCHDYNADJTSNUM=8,  
MainBcchSdcchNum=8;  
  
SET GCELLPSCHM: IDTYPE=BYID, CELLID=1, MAXPDCHRATE=100,  
DYNCHTRANRESLEV=0;
```

Using the MAE-Deployment

NOTE

When configuring the Dynamic SDCCH Adjustment feature on the MAE-Deployment, you must perform a single configuration first, and then perform batch modifications if required. You must perform a single configuration for a parameter before batch modifications of the parameter. You are advised to perform batch modifications before logging out of the parameter setting interface.

Step 1 Configure a single object on the MAE-Deployment. (MAE-Deployment single configuration)

Set parameters on the MAE-Deployment configuration interface according to the operation sequence described in [Table 6-2](#).

For detailed operations on the MAE-Deployment, see Feature Configuration Using the MAE-Deployment.

Step 2 (Optional) Modify objects in batches on the MAE-Deployment. (MAE-Deployment batch modification center)

To modify objects, such as BSCs, BTSs, cells, and TRXs, in batches, click the  icon on the MAE-Deployment configuration interface to start the batch modification wizard.

For instructions on how to perform a batch modification, click  on the wizard interface to obtain online help.

----End

Table 6-2 Configuring parameters on the MAE-Deployment

SN	Managed Object (MO)	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
1	GCELLBASICPARA	BSC6900/BSC6910	SDCCH Dynamic Allocation Allowed	<i>SDDYN</i>	Yes

SN	Managed Object (MO)	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
		BSC6900/ BSC6910	Dynamic PDCH Conversion to SDCCH Allowed	<i>PDCH2SDEN</i>	
2	GCELLCHM GBASIC	BSC6900/ BSC6910	Idle SDCCH Threshold	<i>IDLES DTHRES</i>	Yes
		BSC6900/ BSC6910	Cell SDCCH Channel Maximum	<i>CELLMAXSD</i>	
		BSC6900/ BSC6910	TCH Minimum Recovery Time	<i>MINRESTI METCH</i>	
		BSC6900/ BSC6910	Num of TSs Convertible to SDCCHs per TRX	<i>SDCCHDY NADJTSN UM</i>	
		BSC6900/ BSC6910	Number of TSs Convertible to SDCCHs on BCCH TRX	<i>MainBcch SdcchNu m</i>	
3	GCELLPSC HM	BSC6900/ BSC6910	Maximum Rate Threshold of PDCHs in a Cell	<i>MAXPDC HRATE</i>	Yes
		BSC6900/ BSC6910	Reservation Threshold of Dynamic Channel Conversion	<i>DYNCHTR ANRESLE V</i>	

6.5.3 Activation Observation

NOTE

Monitor the channel status by referring to section "Monitoring Channel Status" in *BSC6900 GSM LMT User Guide* or *BSC6910 GSM LMT User Guide*.

Two TRXs are configured. For the BCCH TRX, timeslot 0 is configured as the BCCH, timeslot 1 is configured as the SDCCH, and timeslots 2 through 7 are configured TCHFs whose **PDCH Channel Priority Type** is set to **EGPRSNORCH(EGPRS Normal Channel)**. For the other TRX, timeslots 1 through 7 are blocked, and only timeslot 0 is not blocked.

This section describes configuration only to enable this feature to take effect as soon as possible. Commercial networks must be configured according to the network plan. After this feature is verified, the configuration must be restored to that before the verification.

- Step 1** On the BSC LMT, run the **SET GCELLCHMGBASIC** command with its parameters set as follows:
- Set **Idle SDCCH Threshold** to 17.
 - Set **Cell SDCCH Channel Maximum** to 100.
 - Set **TCH Minimum Recovery Time** to 60.
 - Set **Num of TSs Convertible to SDCCHs per TRX** to 8.
 - Set **Number of TSs Convertible to SDCCHs on BCCH TRX** to 8.
- Step 2** Run the **SET GCELLPSCHM** command with **Maximum Rate Threshold of PDCHs in a Cell** set to 100 and **Reservation Threshold of Dynamic Channel Conversion** set to 0.
- Step 3** Use an MS to initiate a CS call in the cell.
- Expected result: A TCH is converted into an SDCCH.
- Step 4** Release the CS call. Then, use multiple MSs to simultaneously initiate PS services in the cell.
- Expected result: All full-rate traffic channels (TCHFs) on the BCCH TRX are converted into PDCHs.
- Step 5** Use an MS to initiate a CS call again in the cell.
- Expected result: A TCH is converted into an SDCCH, and a PDCH is simultaneously converted into a TCH. Alternatively, a TCH is occupied by a CS call, and a PDCH is simultaneously converted into an SDCCH.
- Step 6** Enable the cell to remain in the idle state for two minutes. The minimum time required for converting SDCCHs back to TCHs is 120 seconds.
- Expected result: One of the two SDCCHs is converted back to a TCH.
- End

6.5.4 Deactivation

Using MML Commands

On the BSC LMT, run the **SET GCELLBASICPARA** command with **SDCCH Dynamic Allocation Allowed** and **Dynamic PDCH Conversion to SDCCH Allowed** set to **NO(No)**.

MML Command Examples

SET GCELLBASICPARA: IDTYPE=BYID, CELLID=1, SDDYN=NO, PDCH2SDEN=NO;

Using the MAE-Deployment

For detailed operations on the MAE-Deployment, see Feature Configuration Using the MAE-Deployment.

6.6 Deploying Dynamic TCH HRFR Conversion

6.6.1 Deployment Requirements

Table 6-3 Deployment requirements

Aspect	Requirement
Related features	See 4 Related Features .
BSC	None
BTS	None
GSM networking	None
MS	None
MSC	None
License	The license controlling this feature has been activated. For details on how to activate the license, see <i>License Management Feature Parameter Description</i> . For details about license items, see <i>License Control Item Description</i> .
Others	None

6.6.2 Dynamic TCH HR/FR Conversion During Channel Allocation

6.6.2.1 Data Preparation

[Table 6-4](#) lists the data to be prepared for deploying this feature.

Table 6-4 Data preparation

Parameter Name	Parameter ID	NE	Setting Notes	Data Source
TCH Rate Adjust Allow	<i>TCHAJ FLAG (BSC69 00, BSC69 10)</i>	BSC69 00/ BSC69 10	Set this parameter to YES(Yes) .	Network plan
TCH Traffic Busy Threshold	<i>TCHBUSYTHRES (BSC69 00, BSC69 10)</i>	BSC69 00/ BSC69 10	The default value is recommended. To increase the proportion of MSs on TCHs, set this parameter to a small value. To reduce the proportion of MSs on TCHs, set this parameter to a large value.	Network plan
Load of UL-OL Cells Rate Select Allowed	<i>HRIULDRATESELECTION (BSC69 00, BSC69 10)</i>	BSC69 00/ BSC69 10	This parameter is required for concentric cells, including co-BCCH cells.	Network plan
Tch Traffic Busy Overlay Threshold	<i>TCHTRICBUSYOVERLAYTHRES (BSC69 00, BSC69 10)</i>	BSC69 00/ BSC69 10	The default value is recommended. To increase the proportion of MSs on TCHs in the overlaid subcell, set this parameter to a small value. To reduce the proportion of MSs on TCHs in the overlaid subcell, set this parameter to a large value.	Network plan
Tch Traffic Busy Underlay Threshold	<i>TCHTRIBUSYUNDERLAYTHRES (BSC69 00, BSC69 10)</i>	BSC69 00/ BSC69 10	The default value is recommended. To increase the proportion of MSs on TCHs in the underlay subcell, set this parameter to a small value. To reduce the proportion of MSs on TCHs in the underlay subcell, set this parameter to a large value.	Network plan

Parameter Name	Parameter ID	NE	Setting Notes	Data Source
AMR TCH/H Prior Cell Load Threshold	AMRTC HHPRI ORLOAD (BSC6900, BSC6910)	BSC6900/ BSC6910	The default value is recommended. This parameter is required if the cell enables the AMR function. To increase the proportion of MSs on AMR TCHs, set this parameter to a small value. To reduce the proportion of MSs on AMR TCHs, set this parameter to a large value. NOTE - This parameter is valid only if AMRTCHHPRIORALLOW (BSC6900, BSC6910) is set to ON(On). - This parameter is invalid for concentric cells if AMRLOADOPTEN (BSC6900, BSC6910) and HRIUOLDRATESELALLOW (BSC6900, BSC6910) are set to YES(Yes) at the same time.	Network plan
AMR Load Calculation Optimization	AMRLOADOPTEN (BSC6900, BSC6910)	BSC6900/ BSC6910	You are advised to set this parameter to YES(Yes) for a concentric cell. For AMR calls in a concentric cell, set this parameter to YES(Yes) if the proportion of MSs on TCHs needs to be accurately controlled in the overlaid and underlaid subcells. NOTE This parameter is valid only if AMRTCHHPRIORALLOW (BSC6900, BSC6910) is set to ON(On).	Network plan
Threshold for OL Load with AMR HR Preferred	INNAMRTCHHPRIORLOAD (BSC6900, BSC6910)	BSC6900/ BSC6910	The default value is recommended. To increase the proportion of MSs on TCHs in the overlaid subcell for AMR calls, set this parameter to a small value. To reduce the proportion of MSs on TCHs in the overlaid subcell, set this parameter to a large value. NOTE This parameter is valid only if AMRLOADOPTEN (BSC6900, BSC6910) and HRIUOLDRATESELALLOW (BSC6900, BSC6910) are set to YES(Yes), and AMRTCHHPRIORALLOW (BSC6900, BSC6910) is set to ON(On).	Network plan

Parameter Name	Parameter ID	NE	Setting Notes	Data Source
Threshold for UL Load with AMR HR Preferred	<i>OUTA MRTC HHPRI ORLOAD (BSC6900, BSC6910)</i>	BSC6900/ BSC6910	The default value is recommended. To increase the proportion of MSs on TCHHs in the underlaid subcell for AMR calls, set this parameter to a small value. To reduce the proportion of MSs on TCHHs in the underlaid subcell, set this parameter to a large value. NOTE This parameter is valid only if <i>AMRLOADOPTEN (BSC6900, BSC6910)</i> and <i>HRIUOLDRATESELALLOW (BSC6900, BSC6910)</i> are set to YES(Yes), and <i>AMRTCHHPRIORALLOW (BSC6900, BSC6910)</i> is set to ON(On).	Network plan
TCHH Allocation Considering Signal Level	<i>JudgeRxLevWhenAssignHr (BSC6900, BSC6910)</i>	BSC6900/ BSC6910	You are advised to set this parameter to OFF(Off). If this parameter is set to ON(On), you need to consider the signal level when assigning the TCHHs. If the signal level is too low, the TCHF are assigned. If this parameter is set to OFF(Off), the signal level is not considered.	Network plan

Parameter Name	Parameter ID	NE	Setting Notes	Data Source
Level Threshold for TCHH Allocation	MinRxLevWhenAssignHr (BSC6900, BSC6910)	BSC6900/ BSC6910	The default value is recommended. This parameter specifies the lowest signal level threshold for assigning AMR calls the TCHHs. The TCHF are assigned to MSs if the signal level is lower than the parameter value, and otherwise the TCHHs are assigned to MSs. This parameter is valid only if JudgeRxLevWhenAssignHr (BSC6900, BSC6910) is set to ON(On). You are advised to set this parameter to a value smaller than the maximum transmit power of TRXs in a cell minus the value of AMRFULLTOHALFHO-PATHTHRESH (BSC6900, BSC6910). For non-AMR calls, the decision principle is as follows: If NOAMRFULLTOHALFHOALLOW (BSC6900, BSC6910) is set to NO(No), TCHF are preferentially assigned if the RX signal level is lower than the parameter value plus 5. Otherwise, TCHHs are preferentially assigned. If NOAMRFULLTOHALFHOALLOW (BSC6900, BSC6910) is set to YES(Yes), TCHF are preferentially assigned if the RX signal level is lower than the value calculated using the following formula: Value of this parameter + AMRFULLTOHALFHO-PATHTHRESH (BSC6900, BSC6910) – NOAMRFULLTOHALFHOPATHTHRESH (BSC6900, BSC6910) Otherwise, TCHHs are preferentially assigned.	Network plan

6.6.2.2 Activation

Using MML Commands

On the BSC LMT, perform the following steps:

Step 1 Run the **SET GCELLCHMGAD** command to set the threshold for dynamic adjustment between TCHHs and TCHF.

- For non-AMR calls in a common cell, set **TCH Traffic Busy Threshold** to an appropriate value based on the actual requirement. For non-AMR calls in a

concentric cell whose **Load of UL-OL Cells Rate Select Allowed** is set to **YES(Yes)**, set **Tch Traffic Busy Overlay Threshold** and **Tch Traffic Busy Underlay Threshold** to appropriate values.

- For AMR calls, set **AMR TCH/H Prior Allowed** to **ON(On)**, and set other parameters as follows:
 - For common cells, set **AMR TCH/H Prior Cell Load Threshold** to an appropriate value based on the actual requirement.
 - For concentric cells, if either **Load of UL-OL Cells Rate Select Allowed** or **AMR Load Calculation Optimization** is set to **NO(No)**, set **AMR TCH/H Prior Cell Load Threshold** to an appropriate value based on the actual requirement. If **Load of UL-OL Cells Rate Select Allowed** and **AMR Load Calculation Optimization** are both set to **YES(Yes)**, set **Threshold for OL Load with AMR HR Preferred** for overlaid subcells and **Threshold for UL Load with AMR HR Preferred** for underlaid subcells to appropriate values.
- When the BSC preferentially assigns TCHHs, and **TCHH Allocation Considering Signal Level** is set to **ON(On)**, the BSC takes into account the RX signal level in addition to the traffic volume. If the RX signal level is less than the value for **Level Threshold for TCHH Allocation**, TCHF are preferentially assigned. If the RX signal level is greater than or equal to the value for **Level Threshold for TCHH Allocation**, TCHHs are preferentially assigned.

NOTE

When an MS is being connected to the target cell during an inter-cell handover, the **TCHH Allocation Considering Signal Level** parameter takes effect only if all the following conditions are met:

- The A interface uses non-IP transmission.
- The TFO feature has been disabled.
- The **Speech Ver Change Optimize in Handover** parameter has been set to **OFF(Off)**.

Step 2 Run the **SET GTRXDEV** command for each TRX you want to configure. In this step, set **TCH Rate Adjust Allow** of the TRX to **YES(Yes)**.

----End

MML Command Examples

```
SET GTRXDEV: TRXID=0, TCHAJFLAG=YES;
```

```
SET GTRXDEV: TRXID=1, TCHAJFLAG=YES;
```

```
SET GTRXDEV: TRXID=2, TCHAJFLAG=YES;
```

```
SET GCELLCHMGAD: IDTYPE=BYID, CELLID=0, TCHBUSYTHRES=60,  
JudgeRxLevWhenAssignHr=ON, MinRxLevWhenAssignHr=15;
```

Using the MAE-Deployment (Method 1)

 NOTE

When configuring the Dynamic Adjustment Between FR and HR feature on the MAE-Deployment, you must perform a single configuration first, and then perform batch modifications if required.

You must perform a single configuration for a parameter before batch modifications of the parameter. You are advised to perform batch modifications before logging out of the parameter setting interface.

Step 1 Configure a single object on the MAE-Deployment. (MAE-Deployment single configuration)

Set parameters on the MAE-Deployment configuration interface according to the operation sequence described in [Table 6-5](#).

For detailed operations on the MAE-Deployment, see Feature Configuration Using the MAE-Deployment.

Step 2 (Optional) Modify objects in batches on the MAE-Deployment. (MAE-Deployment batch modification center)

To modify objects, such as BSCs, BTSs, cells, and TRXs, in batches, click the  icon on the MAE-Deployment configuration interface to start the batch modification wizard.

For instructions on how to perform a batch modification, click  on the wizard interface to obtain online help.

----End

Table 6-5 Configuring parameters on the MAE-Deployment

SN	Managed Object (MO)	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
1	GTRXDEV	BSC6900/ BSC6910	TCH Rate Adjust Allow	<i>TCHAJFLAG</i>	Yes
2	GCELLCHMGAD	BSC6900/ BSC6910	TCH Traffic Busy Threshold	<i>TCHBUSYTHRES</i>	Yes
		BSC6900/ BSC6910	Load of UL-OL Cells Rate Select Allowed	<i>HRIUOLDR ATESELAL LOW</i>	

SN	Managed Object (MO)	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
		BSC6900/ BSC6910	Tch Traffic Busy Overlay Threshold	<i>TCHTRICBUSYOVERLAYTHR</i>	
		BSC6900/ BSC6910	Tch Traffic Busy Underlay Threshold	<i>TCHTRIBUSYUNDERLAYTHR</i>	
		BSC6900/ BSC6910	AMR TCH/H Prior Cell Load Threshold	<i>AMRTCHHPRIORLOAD</i>	
		BSC6900/ BSC6910	AMR Load Calculation Optimization	<i>AMRLOADOPTEN</i>	
		BSC6900/ BSC6910	Threshold for OL Load with AMR HR Preferred	<i>INNAMRTCHHPRIORLOAD</i>	
		BSC6900/ BSC6910	Threshold for UL Load with AMR HR Preferred	<i>OUTAMRTCHHPRIORLOAD</i>	
		BSC6900/ BSC6910	TCHH Allocation Considering Signal Level	<i>JudgeRxLevelWhenAssignHr</i>	
		BSC6900/ BSC6910	Level Threshold for TCHH Allocation	<i>MinRxLevelWhenAssignHr</i>	

Using the MAE-Deployment (Method 2)

This feature can be batch activated using the MAE-Deployment. For detailed operations, see the following section in the MAE-Deployment product documentation or online help: MAE-Deployment Operation and Maintenance > MAE-Deployment Guidelines > Enhanced Feature Management > Feature Operation and Maintenance.

6.6.2.3 Activation Observation

Assume that non-AMR calls are performed in a common cell. Only the traffic volume, instead of the RX signal level, is considered in assignment of TCHHs. The observation procedure is as follows:

Step 1 Monitor channel status.

Step 2 On the BSC LMT, run the **SET GCELLCHMGAD** command to specify the corresponding cell. In this step, set **TCH Traffic Busy Threshold** to **0** to ensure that new calls are preferentially assigned TCHHs.

Step 3 Use an MS to initiate a call in the cell.

Expected result: The call is successfully initiated, and the communication is normal. A TCHF is converted into two TCHHs and are occupied by MSs.

Step 4 Terminate the call.

Expected result: The two TCHHs are not converted into a TCHF.

Step 5 Run the **SET GTRXCHANADMSTAT** command to block all TCHFs in the cell. In this step, set **Administrative State** to **Lock(Lock)**.

Step 6 Run the **SET GCELLCHMGAD** command with **TCH Traffic Busy Threshold** set to **100** to ensure that new calls are preferentially assigned TCHFs.

NOTE

In [Step 6](#), if the calling party and the called party camp on the same cell, reserve a TCHF to ensure successful setup of the call.

Step 7 Use an MS to initiate a call in the cell.

Expected result: The call is successfully initiated, and the communication is normal. The MS occupies a TCHF.

----End

Assume that non-AMR calls are performed in a common cell. The traffic volume and RX signal level are considered in assignment of TCHHs. The observation procedure is as follows:

Step 1 Monitor channel status.

Step 2 Run the **SET GCELLCHMGAD** command to specify the corresponding cell. In this step, set **TCH Traffic Busy Threshold** and **Level Threshold for TCHH Allocation** to **0**.

Step 3 Use an MS to initiate a call in the cell.

Expected result: The call is successfully initiated, and the communication is normal. The MS occupies a TCHH.

Step 4 Terminate the call.

Expected result: The two TCHHs are not converted into a TCHF.

Step 5 Run the **SET GCELLCHMGAD** command with **Level Threshold for TCHH Allocation** of the corresponding cell set to **63**.

Expected result: The call is successfully initiated, and the communication is normal. The MS occupies a TCHF.

----End

6.6.2.4 Deactivation

Using MML Commands

On the BSC LMT, run the **SET GTRXDEV** command for each TRX you have configured. In this step, set **TCH Rate Adjust Allow** of the TRX to **NO(No)**.

MML Command Examples

SET GTRXDEV: TRXID=0, TCHAJFLAG=NO;

SET GTRXDEV: TRXID=1, TCHAJFLAG=NO;

SET GTRXDEV: TRXID=2, TCHAJFLAG=NO;

Using the MAE-Deployment

The method of feature deactivation using the MAE-Deployment is the same as that of feature activation using the MAE-Deployment. For detailed operations, see [Using the MAE-Deployment \(Method 1\)](#) and [Using the MAE-Deployment \(Method 2\)](#).

6.6.3 Dynamic TCH HR/FR Conversion During a Call

6.6.3.1 Activation

Using MML Commands (Handover Algorithm I)

On the BSC LMT, perform the following steps:

Step 1 Run the BSC MML command **SET GCELLHOBASIC** with **Intracell F-H HO Allowed** set to **HOBASECAPANDQUAL**, **Current HO Control Algorithm** set to **HOALGORITHM1**, and both **AMR F-H Ho Allowed** and **Non-AMR F-H Ho Allowed** set to **YES(Yes)**.

Step 2 Configure FR/HR call conversion.

- Run the BSC MML command **SET GCELLCCACCESS** with **FULL_RATE_VER1(Full-rate VER 1)**, **FULL_RATE_VER2(Full-rate VER 2)**, **FULL_RATE_VER3(Full-rate VER 3)**, **HALF_RATE_VER1(Half-rate VER 1)**,

HALF_RATE_VER3(Half-rate VER 3), and **FULL_RATE_VER5**(Full-rate VER 5) selected under **Speech Version**.

----End

MML Command Examples (Handover Algorithm I)

//Activating Dynamic HR/FR Adaptation

SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM1,
INTRACELLFHHOEN=HOBASECAPANDQUAL, AMRFULLTOHALFHOALLOW=YES,
NOAMRFULLTOHALFHOALLOW=YES;

SET GCELLCCACCESS: IDTYPE=BYID, CELLID=0,
VOICEVER=FULL_RATE_VER1-1&FULL_RATE_VER2-1&FULL_RATE_VER3-1&HALF_R
ATE_VER1-1&HALF_RATE_VER3-1&FULL_RATE_VER5-1;

Using MML Commands (Handover Algorithm II)

On the BSC LMT, perform the following steps:

Step 1 Run the **SET GCELLHOBASIC** command with its parameters set as follows:

Set **Intracell HO Allowed** to **YES(Yes)**.

Set **Current HO Control Algorithm** to **HOALGORITHM2**(Handover algorithm II).

Set both **AMR F-H Ho Allowed** and **Non-AMR F-H Ho Allowed** to **YES(Yes)**.

Step 2 Configure FR/HR call conversion.

- Run the **SET GCELLCCACCESS** command with **FULL_RATE_VER1**(Full-rate VER 1), **FULL_RATE_VER2**(Full-rate VER 2), **FULL_RATE_VER3**(Full-rate VER 3), **HALF_RATE_VER1**(Half-rate VER 1), **HALF_RATE_VER3**(Half-rate VER 3), and **FULL_RATE_VER5**(Full-rate VER 5) selected under **Speech Version**.

----End

MML Command Examples (Handover Algorithm II)

//Activating Dynamic HR/FR Adaptation

SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM2,
INTRACELLHOEN=YES, AMRFULLTOHALFHOALLOW=YES,
NOAMRFULLTOHALFHOALLOW=YES;

SET GCELLCCACCESS: IDTYPE=BYID, CELLID=0,
VOICEVER=FULL_RATE_VER1-1&FULL_RATE_VER2-1&FULL_RATE_VER3-1&HALF_R
ATE_VER1-1&HALF_RATE_VER3-1&FULL_RATE_VER5-1;

Using the MAE-Deployment (Method 1)

NOTE

When configuring the Dynamic HR/FR Adaptation feature on the MAE-Deployment, you must perform a single configuration first, and then perform batch modifications if required. You must perform a single configuration for a parameter before batch modifications of the parameter. You are advised to perform batch modifications before logging out of the parameter setting interface.

Step 1 Configure a single object on the MAE-Deployment. (MAE-Deployment single configuration)

Set parameters on the MAE-Deployment configuration interface according to the operation sequence described in [Table 6-6](#).

For detailed operations on the MAE-Deployment, see Feature Configuration Using the MAE-Deployment.

Step 2 (Optional) Modify objects in batches on the MAE-Deployment. (MAE-Deployment batch modification center)

To modify objects, such as BSCs, BTSs, cells, and TRXs, in batches, click the  icon on the MAE-Deployment configuration interface to start the batch modification wizard.

For instructions on how to perform a batch modification, click  on the wizard interface to obtain online help.

----End

Table 6-6 Configuring parameters on the MAE-Deployment

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
1	GCELLHOBASIC	BSC6900/BSC6910	Intracellular HO Allowed	<i>INTRACELLHOEN</i>	Yes
			Current HO Control Algorithm	<i>HOCTRLSWITCH</i>	

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
			AMR F-H Ho Allowed	AMRFULLTOHALFH OALLOW	
			Non-AMR F-H Ho Allowed	NOAMRFULLTOHAL FHOALLOW	
2	GCELLCCACCESS NOTE Configure this MO to convert FR calls to HR calls.	BSC6900/ BSC6910	Speech Version	VOICEVER	Yes
3	GCELLHOBASIC NOTE Configure this MO to convert FR calls to HR calls.	BSC6900/ BSC6910	AMR F-H Ho Qual. Threshold	AMRFULLTOHALFH OQUALTHRESH	Yes
			Non-AMR F-H Ho Pathloss Threshold	NOAMRFULLTOHAL FHOPATHTHRESH	
			Non-AMR F-H Ho ATCB Threshold	NOAMRFULLTOHAL FHOATCBTHRESH	

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
4	GCELLCCAC CESS NOTE Configure this MO to convert HR calls to FR calls.	BSC6900/ BSC6910	Speech Version	VOICEVER	Yes
5	GCELLCHM GAD NOTE Configure this MO to convert HR calls to FR calls.	BSC6900/ BSC6910	AMR TCH/H Prior Allowed	AMRTCHHPRIORAL LOW	Yes
			AMR TCH/H Prior Cell Load Threshold	AMRTCHHPRIORLO AD	
			TCH Traffic Busy Threshold	TCHBUSYTHRES	
6	GCELLHOB ASIC NOTE Configure this MO to convert HR calls to FR calls.	BSC6900/ BSC6910	AMR H-F Traffic Threshold	AMRHALFTOFULLH OTHRESH	Yes
			AMR H-F Ho Pathloss Threshold	AMRHALFTOFULLH OPATHTHRESH	

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
			Non-AMR H-F Ho ATCB Threshold	<i>NOAMRHALFTOFUL LHOATCBTHRESH</i>	

Using the MAE-Deployment (Method 2)

This feature can be batch activated using the MAE-Deployment. For detailed operations, see the following section in the MAE-Deployment product documentation or online help: MAE-Deployment Operation and Maintenance > MAE-Deployment Guidelines > Enhanced Feature Management > Feature Operation and Maintenance.

6.6.3.2 Activation Observation

Converting FR calls to HR calls

On the BSC LMT, perform the following steps:

- Step 1** Trace BSSAP messages over the A interface and RSL messages over the Abis interface in a cell.
- Step 2** Use an MS to make a call in the cell.
- Step 3** View the BSSAP messages traced over the A interface and the RSL messages traced over the Abis interface.

Expected result: The call has been set up and the voice quality is good. During the call setup, the FAMR speech version is used. During the call, the speech version changes to HAMR. The Handover Detection message is traced over the Abis interface and the Handover Complete message is traced over the Um interface.

NOTE

If **Assignment Procedure in Intra-Cell HO Allowed** in the **SET GCELLSOFT** command is set to **YES(Yes)**, the Assignment Complete message can be traced. If this parameter is set to **NO(No)**, the Handover Complete message can be traced.

----End

Converting HR calls to FR calls

On the BSC LMT, perform the following steps:

- Step 1** Trace BSSAP messages over the A interface and RSL messages over the Abis interface in a cell.
- Step 2** Use an MS to make a call in the cell.
- Step 3** View the BSSAP messages traced over the A interface and the RSL messages traced over the Abis interface.

Expected result: The call has been set up and the voice quality is good. During the call setup, the HAMR speech version is used. During the call, the speech version changes to FAMR. The Handover Detection message is traced over the Abis interface and the Handover Complete message is traced over the Um interface.

 NOTE

If **Assignment Procedure in Intra-Cell HO Allowed** in the **SET GCELLSOFT** command is set to **YES(Yes)**, the Assignment Complete message can be traced. If this parameter is set to **NO(No)**, the Handover Complete message can be traced.

----End

6.6.3.3 Deactivation

Using MML Commands (Handover Algorithm I)

On the BSC LMT, perform the following steps:

- Step 1** If handover algorithm I is used, run the BSC MML command **SET GCELLHOBASIC** with **Intracell F-H HO Allowed** set to **NO(No)**.
- Step 2** Run the BSC MML command **LST GCELLHOBASIC** to query the value of **Intracell F-H HO Allowed**.

Expected result: The value of **Intracell F-H HO Allowed** is **No**.

----End

MML Command Examples (Handover Algorithm I)

//Deactivating Dynamic HR/FR Adaptation

SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM1,
INTRACELLFHNOEN=NO;

Using MML Commands (Handover Algorithm II)

On the BSC LMT, perform the following steps:

- Step 1** Run the **SET GCELLHOBASIC** command with **AMR F-H Ho Allowed** set to **NO(No)**.
- Step 2** Run the **LST GCELLHOBASIC** command to query the value of **AMR F-H Ho Allowed**.

Expected result: The value of **AMR F-H Ho Allowed** is **No**.

- Step 3** Run the **SET GCELLHOBASIC** command with **Non-AMR F-H Ho Allowed** set to **NO(No)**.

Step 4 Run the **LST GCELLHOBASIC** command to query the value of **Non-AMR F-H Ho Allowed**.

Expected result: The value of **Non-AMR F-H Ho Allowed** is **No**.

----End

MML Command Examples (Handover Algorithm II)

//Deactivating Dynamic HR/FR Adaptation

SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM2,
AMRFULLTOHALFHOALLOW=NO, NOAMRFULLTOHALFHOALLOW=NO;

Using the MAE-Deployment

For detailed operations on the MAE-Deployment, see Feature Configuration Using the MAE-Deployment.

6.7 Deploying Dynamic Multiple CCCH

6.7.1 Deployment Requirements

Table 6-7 Deployment requirements

Aspect	Requirement
Related features	See 4 Related Features .
BSC	None
BTS	Both 3900 series base stations (excluding BTS3900B and BTS3900E) and 5900 series base stations support this feature.
GSM networking	None
MS	None
MSC	None
License	The license controlling this feature has been activated. For details on how to activate the license, see <i>License Management Feature Parameter Description</i> . For details about license items, see <i>License Control Item Description</i> .
Others	None

6.7.2 Precautions

- If timeslot 0 of the BCCH TRX is combined BCCH or BCCH+CBCH, the Dynamic Multiple CCCH feature cannot be enabled for the BCCH TRX.
- If the **Dynamic Channel Adjustment Policy** parameter is set to **OFF(Off)** when you enable the Dynamic Multiple CCCH feature, a rollback cannot be performed on the extended CCCHs dynamically converted.
- You are advised to enable or disable this feature in all the cells under the same location area or BSC.

NOTE

The paging mode used by the Huawei BSS can be location area code (LAC) paging or BSC paging.

- Configure the **BCCH TRX RSL Multiplexing Mode** parameter as follows:
 - TMSI paging: When **Maximum Number of Dynamic CCCHs** plus the number of static BCHs is 1, 2, or 3, you are advised to set **BCCH TRX RSL Multiplexing Mode** to **MODE2_1**, **MODE1_1**, and **MODE1_1**, respectively.
 - IMSI paging: When **Maximum Number of Dynamic CCCHs** plus the number of static BCHs is 1, 2, or 3, you are advised to set **BCCH TRX RSL Multiplexing Mode** to **MODE4_1**, **MODE2_1**, and **MODE1_1**, respectively.

NOTE

A large value of **Maximum Number of Dynamic CCCHs** represents a high Um interface capability. To prevent message delivery congestion over the Abis interface, you need to increase the RSL bandwidth of BCCHs and set **BCCH TRX RSL Multiplexing Mode** by referring to the preceding suggestions.

- To support the Dynamic Multiple CCCH feature, configure SDCCHs, static PDCHs, or channels reserved for VIP users that involve in the baseband frequency hopping in the cell by complying with the following rules:
 - If **Maximum Number of Dynamic CCCHs** is set to **1**, the channels cannot be configured in timeslot 2.
 - If **Maximum Number of Dynamic CCCHs** is set to **2**, the channels cannot be configured in timeslot 2 or 4.
 - If **Maximum Number of Dynamic CCCHs** is set to **3**, the channels cannot be configured in timeslot 2, 4, or 6.
- For emergency calls that must be guaranteed, set the **EMCPRILV (BSC6900, BSC6910)** to a value other than 15. Otherwise, if the emergency call is assigned with a channel that is to be converted to a CCCH, the emergency call will be handed over to another channel. If the handover fails, the call cannot return to the original channel. As a result, the emergency call drops.

6.7.3 Data Preparation

Table 6-8 lists the data to be prepared for deploying this feature.

Table 6-8 Data preparation

Parameter Name	Parameter ID	NE	Setting Notes	Data Source
Dynamic Multiple CCCHs	<i>DynCCCHSwitch</i> (BSC6900, BSC6910)	BSC6900/ BSC6910	If static BCHs are configured in a cell, delete static BCHs and enable the Dynamic Multiple CCCH feature in all cells under the location area or BSC. If over 70% of all cells under a location area or BSC experience paging timeout during peak hours on the live network, disable this feature in all the cells under the location area or BSC. If over 160,000 paging messages are sent in one peak hour in a cell on the live network, enable this feature in all the cells under the location area or BSC.	Internal plan
Dynamic Channel Adjustment Policy	<i>ChannelDynamicAdjustment</i> (BSC6900, BSC6910)	BSC6900/ BSC6910	Whether to enable the dynamic channel adjustment optimization procedure. If this parameter is set to ON(On), the BSC sends the BTS a message indicating dynamic channel adjustment over the RSL. If this parameter is set to OFF(Off), the BSC sends the BTS a message indicating dynamic channel adjustment over the OML. By default, the configuration data sent to the BTS over the OML is static configuration, and a rollback cannot be performed on extended CCCHs dynamically converted. Therefore, you must set this parameter to ON(On) when enabling the Dynamic Multiple CCCH feature.	Default/Recommended Value

Parameter Name	Parameter ID	NE	Setting Notes	Data Source
Dynamic CCCH Addition Threshold	<i>DynCCCHLoadIncrThr</i> <i>(BSC6900, BSC6910)</i>	BSC6900/ BSC6910	Load threshold for dynamically converting a TCH to an extended CCCH. If the CCCH load is greater than the value of this parameter within the duration specified by <i>DynCCCHLoadDuration</i> (<i>BSC6900, BSC6910</i>), a TCH is dynamically converted to an extended CCCH. The larger the value of this parameter, the smaller the number of TCHs dynamically converted to extended CCCHs during peak hours, and the lower the paging efficiency. The smaller the value of this parameter, the larger the number of TCHs dynamically converted to extended CCCHs during off-peak hours, and the lower the resource usage.	Network plan

Parameter Name	Parameter ID	NE	Setting Notes	Data Source
Dynamic CCCH Removal Threshold	<i>DynCCCHLoadRedcTh</i> <i>r</i> <i>(BSC6900, BSC6910)</i>	BSC6900/ BSC6910	Load threshold for dynamically converting an extended CCCH to a TCH. An extended CCCH can be dynamically converted to a TCH if the following conditions are met: The CCCH load is less than the value of this parameter within the duration specified by <i>DynCCCHLoadDuration</i> (<i>BSC6900, BSC6910</i>). The BSC predicts that the message sending capability on the Um interface is sufficient after the number of CCCHs decreases. The interval between two consecutive conversions reaches the value specified by <i>DynCCCHReduceInterval</i> (<i>BSC6900, BSC6910</i>). The larger the value of this parameter, the smaller the number of extended CCCHs dynamically converted to TCHs during peak hours, and the lower the paging efficiency. The smaller the value of this parameter, the larger the number of extended CCCHs dynamically converted to TCHs during off-peak hours, and the lower the resource usage.	Network plan

Parameter Name	Parameter ID	NE	Setting Notes	Data Source
Dynamic CCCH Load Stat. Duration	<i>DynCCCHLoadDuration</i> <i>(BSC6900, BSC6910)</i>	BSC6900/ BSC6910	Duration during which the CCCH load is higher than the value of <i>DynCCCHLoadIncrThr</i> (<i>BSC6900, BSC6910</i>) or lower than the value of <i>DynCCCHLoadRedcThr</i> (<i>BSC6900, BSC6910</i>). After this duration expires, the BTS triggers a dynamic conversion between TCHs and extended CCCHs. The larger the value of this parameter, the longer the time to trigger dynamic conversion between extended CCCHs and TCHs, the smaller the number of dynamic conversions, and the lower the paging efficiency. The smaller the value of this parameter, the shorter the time to trigger dynamic conversion between extended CCCHs and TCHs, the larger the number of unnecessary conversions due to inaccurate determination, and the lower the resource usage.	Network plan

Parameter Name	Parameter ID	NE	Setting Notes	Data Source
Maximum Number of Dynamic CCCHs	<i>MaxDynCCCHNum</i> <i>(BSC6900, BSC6910)</i>	BSC6900/ BSC6910	Maximum number of extended CCCHs that can be dynamically converted in a cell. Each cell has a maximum of three extended CCCHs, including dynamically converted CCCHs and statically configured CCCHs. These CCCHs are carried by timeslots 2, 4, and 6 on the BCCH TRX. When the CCCH load is low, the channels on these timeslots can be occupied by signaling, CS services, or PS services. When the CCCH load is high, the channels on these timeslots are dynamically converted to extended CCCHs to reduce the CCCH load. The larger the value of this parameter, the larger the number of TCHs dynamically converted to CCCHs during peak hours, the smaller the number of available TCHs, and the higher the TCH congestion rate. The smaller the value of this parameter, the smaller the number of TCHs dynamically converted to CCCHs, and the lower the paging efficiency.	Network plan

Parameter Name	Parameter ID	NE	Setting Notes	Data Source
Interval for Dynamically Removing CCCHs	<i>DynCCCHReduceInterval</i> <i>(BSC6900, BSC6910)</i>	BSC6900/ BSC6910	Period during which the next dynamic conversion from extended CCCHs to TCHs is prohibited after a dynamic conversion. Specifying such period prevents frequent dynamic conversions due to CCCH load changes. The larger the value of this parameter, the smaller the number of extended CCCHs dynamically converted to TCHs during off-peak hours, and the lower the resource usage. The smaller the value of this parameter, the larger the number of extended CCCHs dynamically converted to TCHs, the more the number of dynamic conversions due to paging burst, and the lower the paging efficiency. You are advised to set this parameter to a value greater than or equal to 30 minutes.	Network plan

Parameter Name	Parameter ID	NE	Setting Notes	Data Source
Priority of Emergency Call	<i>EMCPRILV</i> <i>(BSC6900, BSC6910)</i>	BSC6900/ BSC6910	Whether the BSC reserves TCHFs for emergency calls and whether emergency calls can preempt TCHFs used by common calls. If the parameter is set to a value ranging from 1 to 14, the BSC reserves TCHFs for emergency calls after receiving the ESTABLISH INDICATION message. If the channel reservation fails and <i>PREEMPTIONPERMIT</i> (<i>BSC6900, BSC6910</i>) is set to ON(ON), the BSC uses the value of this parameter as the user priority (1 is the highest priority and 14 is the lowest one) for emergency calls to preempt TCHFs used by common calls. After successfully preempting the TCHFs, the BSC reserves the TCHFs for emergency calls. When this parameter is set to a value ranging from 1 to 14, the TCH occupied by an emergency call cannot be preempted. The specific handling policies are as follows: 1. The TCHs occupied by emergency calls cannot be preempted. 2. If <i>DynCCCHSwitch</i> (<i>BSC6900, BSC6910</i>) is set to ON(ON), emergency calls on channels corresponding to timeslots 2, 4, and 6 on the BCCH TRX can be forcibly handed over to another channels. If the forcible handover fails, the emergency calls can return to the original channels and will not be released.	Network plan

6.7.4 Activation

Using MML Commands

Step 1 On the BSC LMT, run the **SET GCELLCCCH** command with its parameters set as follows:

- Set **Dynamic Multiple CCCHs** to **ON(ON)**.
- Set **Dynamic CCCH Addition Threshold**, **Dynamic CCCH Removal Threshold**, **Dynamic CCCH Load Stat. Duration**, **Maximum Number of Dynamic CCCHs**, and **Interval for Dynamically Removing CCCHs** to appropriate values.

Step 2 Run the **SET BSCEXSOFTPARA** command with **Dynamic Channel Adjustment Policy** set to **ON(On)**.

----End

NOTE

If the BCCH TRX participates in the baseband frequency hopping (FH), timeslots that do not participate in the frequency hopping vary with the value of **Maximum Number of Dynamic CCCHs**. The specific mapping is as follows:

- Value 1: timeslot 2
- Value 2: timeslots 2 and 4
- Value 3: timeslots 2, 4, and 6

When the cell is in the deactivated state, you can run the SET GTRXCHANHOP command on the BSC LMT to change the FH attributes of timeslots 2, 4, and 6 for BCCH and non-BCCH TRXs.

MML Command Examples

SET GCELLCCCH: IDTYPE=BYID, CELLID=2, DynCCCHSwitch=ON,
MaxDynCCCHNum=2, DynCCCHLoadIncrThr=90, DynCCCHLoadRedcThr=65,
DynCCCHLoadDuration=10, DynCCCHReduceInterval=30;

SET BSCEXSOFTPARA: ChannelDynAdjustOpt=ON;

Using the MAE-Deployment

NOTE

When configuring the Dynamic Multiple CCCH feature on the MAE-Deployment, you must perform a single configuration first, and then perform batch modifications if required.

You must perform a single configuration for a parameter before batch modifications of the parameter. You are advised to perform batch modifications before logging out of the parameter setting interface.

Step 1 Configure a single object on the MAE-Deployment. (MAE-Deployment single configuration)

Set parameters on the MAE-Deployment configuration interface according to the operation sequence described in [Table 6-9](#).

For detailed operations on the MAE-Deployment, see Feature Configuration Using the MAE-Deployment.

Step 2 (Optional) Modify objects in batches on the MAE-Deployment. (MAE-Deployment batch modification center)

To modify objects, such as BSCs, BTSs, cells, and TRXs, in batches, click the  icon on the MAE-Deployment configuration interface to start the batch modification wizard.

For instructions on how to perform a batch modification, click  on the wizard interface to obtain online help.

----End

Table 6-9 Configuring parameters on the MAE-Deployment

SN	Managed Object (MO)	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
1	GCELLCCH	BSC6900/ BSC6910	Dynamic Multiple CCCHs	<i>DynCCCHSwitch</i>	Yes
		BSC6900/ BSC6910	Dynamic CCCH Addition Threshold	<i>DynCCCHLoadIncrThr</i>	
		BSC6900/ BSC6910	Dynamic CCCH Removal Threshold	<i>DynCCCHLoadRedcThr</i>	
		BSC6900/ BSC6910	Dynamic CCCH Load Stat. Duration	<i>DynCCCHLoadDuration</i>	
		BSC6900/ BSC6910	Maximum Number of Dynamic CCCHs	<i>MaxDynCCCHNum</i>	
		BSC6900/ BSC6910	Interval for Dynamically Removing CCCHs	<i>DynCCCHReduceInterval</i>	

SN	Managed Object (MO)	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
2	BSCXSOF TPARA	BSC6900/ BSC6910	Dynamic Channel Adjustment Policy	<i>ChannelDynamicAdjustment</i>	Yes

6.7.5 Activation Observation

Increasing Dynamic CCCHs

- Step 1** On the BSC LMT, run the **SET GCELLCCCH** command with its parameters set as follows:
- Set **Dynamic Multiple CCCHs** to **ON(ON)**.
 - Set **Dynamic CCCH Addition Threshold**, **Dynamic CCCH Removal Threshold**, and **Dynamic CCCH Load Stat. Duration** to **0**, and **Maximum Number of Dynamic CCCHs** to **3**.
- Step 2** Initiate a large number of calls (including originated calls and terminated calls). Ensure that more than 100 paging messages are delivered within a second.
- Step 3** View the traced CS messages on the Abis interface.

NOTE

Trace Abis-interface messages over both the OML and RSL when either of the following conditions are met:

The dynamic CCCH adjustment messages are sent over both the RSL and OML.

The GSM and UMTS Dynamic Spectrum Sharing(GSM) feature is enabled, but the BTS does not report its capability.

In other cases, the BSC traces Abis-interface messages only over the RSLs.

For details about how to trace Abis-interface messages, see "Tracing CS Domain OML Messages on the Abis Interface" and "Tracing CS Domain RSL Messages on the Abis Interface" in *BSC6900 GSM LMT User Guide* or *BSC6910 GSM LMT User Guide*.

- RSL message tracing
If all the following messages are traced, the CCCH dynamic adjustment is successful:
 - Abis Dynamic Ccch Request
 - Abis Dynamic Ccch Request Ack
 - OAM in RSL (change-chan-comb)
 - OAM in RSL (change-chan-comb-ack)
- RSL and OML message tracing
The following messages are traced over the RSL:

- Abis Dynamic Ccch Request
 - Abis Dynamic Ccch Request Ack
- The following messages are traced over the OML:
- Set Channel Attributes
 - Set Channel Attributes Ack

Step 4 Monitor channel status.

The monitoring window on the BSC LMT shows that timeslots 2, 4, and 6 on the BCCH TRX are converted to BCHs in sequence based on the CCCH load. For details about how to monitor the channel status, see "Monitoring Channel Status" in *BSC6900 GSM LMT User Guide* or *BSC6910 GSM LMT User Guide*.

Step 5 Check the related counters.

Check the value of the CELL.CH.AVAIL.NUM.CCCH.AVR counter. If the value of this counter minus the number of static CCCHs is not 0, the TCH has been converted to an extended CCCH.

----End

Decreasing Dynamic CCCHs

Step 1 If the number of added dynamic CCCH is greater than 1, run the **SET GCELLCCCH** command with its parameters set as follows on the BSC LMT:

- Set **Dynamic CCCH Addition Threshold** to **90**.
- Set **Dynamic CCCH Removal Threshold** to **90**.
- Set **Dynamic CCCH Load Stat. Duration** to **0**.
- Set **Interval for Dynamically Removing CCCHs** to **1**.

Step 2 Stop initiating calls.

Step 3 Trace CS messages over the Abis interface.

 NOTE

Trace Abis-interface messages over both the OML and RSL when either of the following conditions is met:

- The dynamic CCCH adjustment messages are sent over both the RSL and OML.
- The GSM and UMTS Dynamic Spectrum Sharing(GSM) feature is enabled, but the BTS does not report its capability.

In other cases, the BSC traces the Abis-interface messages only over RSLs.

For details about how to trace Abis-interface messages, see "Tracing CS Domain OML Messages on the Abis Interface" and "Tracing CS Domain RSL Messages on the Abis Interface" in *BSC6900 GSM LMT User Guide* or *BSC6910 GSM LMT User Guide*.

- RSL message tracing

If all the following messages are traced, the CCCH dynamic adjustment is successful:

- Abis Dynamic Ccch Request
- Abis Dynamic Ccch Request Ack
- OAM in RSL (change-chan-comb)

- OAM in RSL (change-chan-comb-ack)
- RSL and OML message tracing
The following messages are traced over the RSL:
 - Abis Dynamic Ccch Request
 - Abis Dynamic Ccch Request AckThe following messages are traced over the OML:
 - Set Channel Attributes
 - Set Channel Attributes Ack

Step 4 Monitor channel status.

The monitoring window on the BSC LMT shows that timeslots 6, 4, and 2 on the BCCH TRX are converted to TCHs in sequence based on the CCCH load. For details about how to monitor the channel status, see "Monitoring Channel Status" in *BSC6900 GSM LMT User Guide* or *BSC6910 GSM LMT User Guide*.

Step 5 Check the related counters.

Check the value of the CELL.CH.AVAIL.NUM.CCCH.AVR counter. If the value of this counter decreases, the CCCH has been dynamically converted to TCHs.

----End

6.7.6 Deactivation

Using MML Commands

 NOTE

If the BCCH TRX participates in the baseband frequency hopping, deactivate the cell and run the BSC MML command **SET GTRXCHANHOP** command to set timeslots 2, 4, and 6 of the BCCH TRX to participate in the baseband frequency hopping before deactivating this feature.

On the BSC LMT, run the **SET GCELLCCCH** command with **Dynamic Multiple CCCHs** set to **OFF(OFF)**.

MML Command Examples

SET GCELLCCCH: IDTYPE=BYID, CELLID=2, DynCCCHSwitch=OFF;

Using the MAE-Deployment

For detailed operations on the MAE-Deployment, see Feature Configuration Using the MAE-Deployment.

6.8 Performance Optimization

6.8.1 Dynamic SDCCH Conversion

Monitoring

Table 6-10 Monitoring counters

Description	Name
Number of Channel Conversions (TCH-SDCCH)	CELL.CH.CONVERT.NUM.TCH.SD
Number of Channel Conversions (SDCCH-TCH)	CELL.CH.CONVERT.NUM.SD.TCH
Congestion Rate on SDCCH per CELL (due to Busy)	CELL.RATE.SD.CONG

Parameter Optimization

If the counter Congestion Rate on SDCCH per CELL (due to Busy) increases, increase the value of **CELLMAXSD (BSC6900, BSC6910)** accordingly.

If **PDCH2SDEN (BSC6900, BSC6910)** is set to **YES(Yes)**, the value of the K3045:Congestion Rate on TCH (All Channels Busy) counter increases, indicating the cell provides a large number of CS services, and the BSC preferentially converts occupied TCHs to SDCCHs.

If **PDCHtoSDCCHAdjustPrefSw (BSC6900, BSC6910)** is set to **ON(On)**, when the cell has sufficient dynamic PDCHs, occupied TCHs, and no idle TCH, the BSC preferentially converts PDCHs to SDCCHs in a dynamic manner.

6.8.2 Dynamic TCH HR/FR Conversion

Monitoring

Table 6-11 Monitoring counters

Description	Name
Mean Number of Busy TCHs (TCHH)	CELL.CH.BUSY.NUM.HR.TRAF.900.AVR + CELL.CH.BUSY.NUM.HR.TRAF.1800.AVR
Congestion Rate on TCH (All Channels Busy)	CELL.KPI.TCH.CONGESTION.RATE
Channel Conversion Requests (TCHF-TCHH)	CELL.CH.CONVERT.REQ.NUM.FR.TO.HR
Successful Channel Conversions (TCHF-TCHH)	CELL.CH.CONVERT.SUCC.NUM.FR.TO.HR

Description	Name
Channel Conversion Requests (TCHH-TCHF)	CELL.CH.CONVERT.REQ.NUM.HR.FR
Successful Channel Conversions (TCHH-TCHF)	CELL.CH.CONVERT.SUCC.NUM.HR.FR

Parameter Optimization

If the half-rate traffic is light, and the value of Congestion Rate on TCH (All Channels Busy) is large, increase the values of ***TCHBUSYTHRES (BSC6900, BSC6910)*** and ***AMRTCHHPRIORLOAD (BSC6900, BSC6910)*** to increase the proportion of TCHHs in a cell, decreasing the congestion rate.

6.8.3 Dynamic PDCH Conversion

Monitoring

Table 6-12 Monitoring counters

Description	Name
Number of Successful Channel Conversions (PDCH to TCHH)	CELL.CH.CONVERT.SUCC.NUM.PDCH.HR
Number of Successful Channel Conversions (PDCH to TCHF)	CELL.CH.CONVERT.SUCC.NUM.PDCH.FR
Number of Successful Channel Conversions (TCHH to PDCH)	CELL.CH.CONVERT.SUCC.NUM.HR.PDCH
Number of Successful Channel Conversions (TCHF to PDCH)	CELL.CH.CONVERT.SUCC.NUM.FR.PDCH

Parameter Optimization

None

6.8.4 Dynamic Multiple CCCH

Monitoring

Table 6-13 Monitoring counters

Description	Name
Mean Number of Available Channels (Extended CCCH)	CELL.CH.AVAIL.NUM.CCCH.AVR

Parameter Optimization

If there is no static CCCH in the cell and paging expiration or deletion occurs in 70% of cells under the LAC where this cell is located, you are advised to set ***MaxDynCCCHNum (BSC6900, BSC6910)*** to **1** or a larger value for all cells under the LAC.

6.9 Troubleshooting

None

7 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

Base station controller parameter reference

- BSC6900 GSM Parameter Reference: contains the parameters related to the BSC6900 equipment, transport, and radio access functions.
- BSC6910 GSM Parameter Reference: contains the parameters related to the BSC6910 equipment, transport, and radio access functions.

NOTE

You can find the EXCEL files of parameter reference for the software version used on the live network from the "Operation and Maintenance > Configuration Management" node in the *BSC69xx Product Documentation* delivered with that version.

eGBTS Parameter reference

- Node Parameter Reference: contains base station equipment and transport parameters.
- GBTSFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.

NOTE

You can find the EXCEL files of parameter reference for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

1. Open the EXCEL file of parameter reference.
2. On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, GBFD-200201.
3. Click **OK**. All parameters related to the feature are displayed.

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

1. Open the EXCEL file of the used reserved parameter list.
2. On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

8 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

Base station controller performance counter reference

- BSC6900 GSM Performance Counter Reference: contains the counters related to the BSC6900 equipment, transport, and radio access functions.
- BSC6910 GSM Performance Counter Reference: contains the counters related to the BSC6910 equipment, transport, and radio access functions.

NOTE

You can find the EXCEL files of performance counter reference for the software version used on the live network from the "Operation and Maintenance > Performance Management" node in the *BSC69xx GSM Product Documentation* delivered with that version.

eGBTS performance counter reference

- Node Performance Counter Summary: contains base station equipment and transport counters.
- GBTSFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.

NOTE

You can find the EXCEL files of performance counter reference for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

1. Open the EXCEL file of performance counter reference.
2. On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, GBFD-200201.
3. Click **OK**. All counters related to the feature are displayed.

9 Glossary

For the acronyms, abbreviations, terms, and definitions, see the *Glossary*.

10 Reference Documents

1. BSC6900 Feature List
2. BSC6900 Basic Feature Description
3. BSC6900 Optional Feature Description
4. BSC6900 GSM Parameter Reference
5. BSC6900 GSM MML Command Reference
6. BSC6900 Performance Counter Reference